

Image Classification and Recognition

Fine-Grained Recognition, Metric Learning, and Open-Set Recognition

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1. Motivation: recognition beyond the closed-set benchmark
2. Adapting pretrained classifiers: fine-tuning under distribution shift, weight averaging, prompt learning, and training-free adapters
3. Fine-grained recognition: benchmarks, part-aware transformers, foundation features, and taxonomic supervision
4. Deep metric learning: ranking metrics, pair, proxy and margin losses, and a reality check
5. Image retrieval: landmark benchmarks, global descriptors, re-ranking, perceptual similarity, and search at scale
6. Re-identification: protocol, strong baselines, transformer and CLIP-based models, and responsible use
7. Open-set recognition and out-of-distribution detection: definitions, metrics, and scoring methods
8. Unknown classes with modern models: benchmarks, zero-shot detection with CLIP, and category discovery
9. Limitations, summary, and references

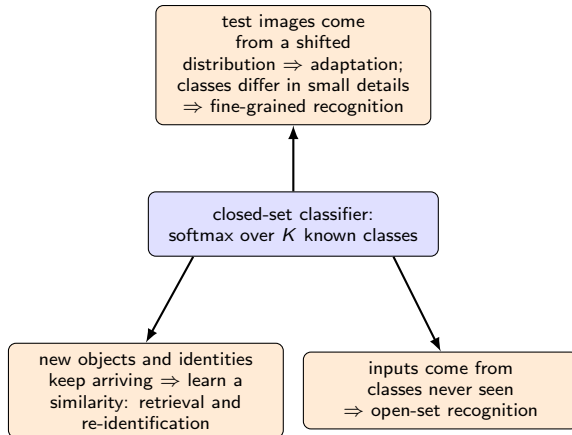
- ▶ **Covered earlier:** CNN and ViT backbones trained on ImageNet; transfer learning and the choice between freezing and fine-tuning; linear and attentive probes for evaluating frozen features.
- ▶ **Covered earlier:** contrastive objectives (InfoNCE, NT-Xent) with a temperature, and the self-distillation recipe behind DINO and DINOv2.
- ▶ **Covered earlier:** CLIP's joint image–text space and zero-shot classification from class-name prompts, including its weakness on fine-grained categories.
- ▶ **Today:** using these encoders for recognition when the assumptions of a closed-set benchmark break. Classifiers are adapted and sharpened, similarities are learned instead of labels, and inputs from classes the model has never seen are rejected.

By the end of this lecture you should be able to:

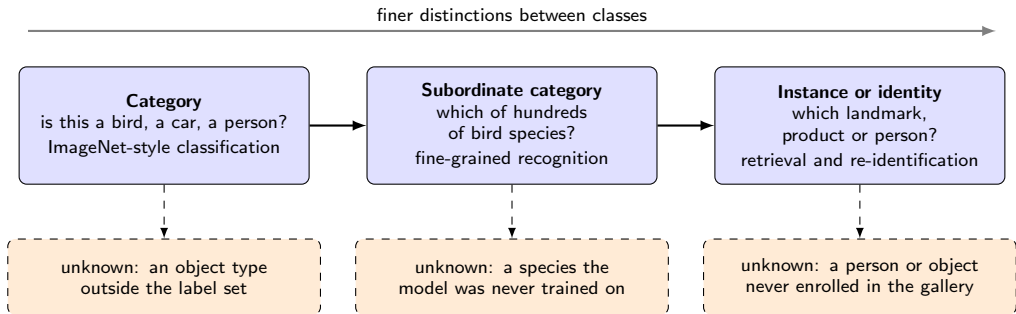
- ▶ **Choose** an adaptation recipe for a pretrained classifier, weighing in-distribution accuracy against accuracy under distribution shift.
- ▶ **Explain** why fine-grained recognition is hard and how part-aware transformers, foundation features and taxonomic labels help.
- ▶ **Derive and compare** pair-, proxy- and margin-based metric-learning losses and the ranking metrics used to judge them.
- ▶ **Design** an image-retrieval pipeline, from global descriptor to re-ranking to approximate nearest-neighbour search.
- ▶ **Describe** the re-identification protocol and the methods behind current results.
- ▶ **Compute and compare** scores that flag inputs from unknown classes, and evaluate them.
- ▶ **Explain** how foundation models detect unknown classes zero-shot and discover new categories in unlabelled data.

Assumed background (all covered earlier): softmax, k -nearest neighbours, k -means, mean average precision, CNN and ViT backbones, contrastive learning, CLIP, DINOv2.

- ▶ A benchmark such as ImageNet fixes three things before training starts:
 - a label set of K classes;
 - test images drawn from the same distribution as the training images;
 - every test image belongs to one of the K classes.
- ▶ A softmax head (covered earlier) builds the first and the last into the model: it spreads a probability of 1 over the same K labels for every input, including an image of something else.
- ▶ Deployed recognition breaks each assumption, and each break needs a different tool.



The three parts of this lecture, one per broken assumption; fine-grained recognition is the closed set at its hardest.

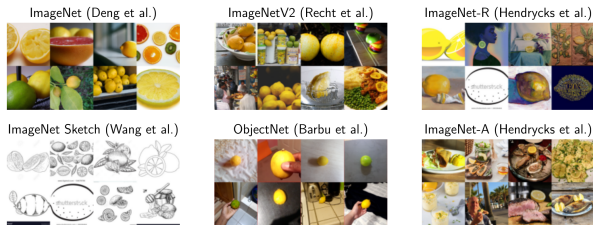


- ▶ Moving right, classes become so numerous and so similar that a fixed softmax head stops being practical: at the instance level every object or person is its own class, and the system compares images instead of predicting a label.
- ▶ The dashed row is present at every level: whatever the granularity, some test inputs belong to no known class.

Question	Output	What is learned	How it is scored
Which of my K classes is this, even when classes differ in small details?	one label	a classifier on adapted pretrained features	top-1 accuracy, in distribution and under distribution shift
Which stored images show the same object, place or person?	a ranked list	an embedding whose distances reflect similarity or identity	how often a correct match appears among the top-ranked results; mean average precision
Is this one of my classes at all?	a label or "unknown"	a classifier, a score of how known the input looks, and a rejection threshold	how well the score separates known from unknown inputs

- ▶ The rows follow the lecture: classification and fine-grained recognition; metric learning, retrieval and re-identification; open-set recognition.
- ▶ Pretrained encoders such as CLIP and DINOv2 (covered earlier) changed all three rows. Each part asks what a frozen encoder already provides and what still has to be trained.

- **Distribution shift** in this section: covariate shift (covered earlier), the same classes in a new image style or source.
- **ID** (in-distribution) accuracy: on the distribution used for fine-tuning. **OOD** (out-of-distribution) accuracy: on shifted test sets with the same classes.
- ImageNet-R holds renditions such as paintings and sculptures; ImageNet-A holds natural images that a ResNet-50 misclassifies.
- Zero-shot CLIP holds up well under shift (covered earlier). Fine-tuning it end-to-end on ImageNet gains 9.6 points there and loses 4.8 points of mean shift accuracy:

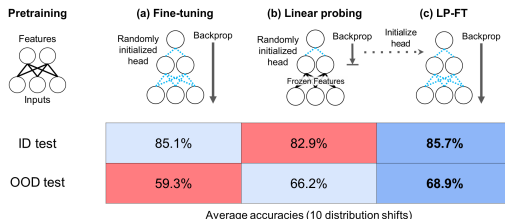


Samples of the class lemon from ImageNet and five shifted test sets. Wortsman et al., "Robust fine-tuning of zero-shot models," CVPR 2022, [arXiv:2109.01903v3](https://arxiv.org/abs/2109.01903v3), Figure 2.

CLIP ViT-L/14@336px	IN	IN-V2	IN-R	IN-Sketch	ObjectNet	IN-A	Mean, 5 shifts
Zero-shot	76.6	70.5	89.0	60.9	69.1	77.7	73.4
Fine-tuned end-to-end	86.2	76.8	79.8	57.9	63.3	65.4	68.6

Top-1 accuracy (%); IN: ImageNet. Wortsman et al., [arXiv:2109.01903v3](https://arxiv.org/abs/2109.01903v3), Table 1.

- ▶ **LP** (linear probing, covered earlier): train a new head on frozen features. **FT** (full fine-tuning): update all weights.
- ▶ Over 10 shift benchmarks, FT wins on average in ID accuracy and LP in OOD accuracy (figure).
- ▶ **Feature distortion**: with a random head, fitting the training data changes the features of training images a lot. In the paper's linear model, inputs orthogonal to the training data keep their old features, which the new head no longer fits.
- ▶ **LP-FT** (LP, then FT): probe the head first, then fine-tune everything from it. On Living-17 (17 animal classes), features move about $20\times$ less than under FT.
- ▶ About the same compute as FT: the probe is cheap, and on ImageNet each stage ran for half the FT epochs.



Kumar et al., "Fine-Tuning can Distort Pretrained Features and Underperform Out-of-Distribution," ICLR 2022, [arXiv:2202.10054v1](https://arxiv.org/abs/2202.10054v1), Figure 1.

ViT-B/16	IN	ImageNet shifts			
		V2	R	Sketch	A
FT	81.7	71.5	52.4	40.5	27.8
LP	79.7	69.7	70.6	46.4	45.7
LP-FT	81.7	71.6	72.9	48.4	49.1

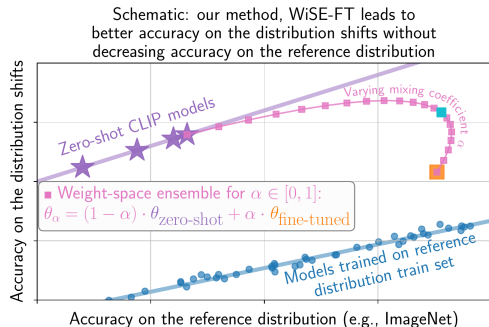
CLIP ViT-B/16 trained on ImageNet, top-1 (%). Same paper, Tables 1 and 2.

- **WiSE-FT** (weight-space ensembles for fine-tuning): fine-tune CLIP end-to-end, starting from its zero-shot head, then interpolate the weights:

$$\theta_{\alpha} = (1 - \alpha) \theta_0 + \alpha \theta_1$$

θ_0 : zero-shot weights, with the head built from the class prompts; θ_1 : fine-tuned weights; $\alpha \in [0, 1]$: mixing coefficient.

- Averaging two networks trained from random initialisations is no better than a random classifier. These two share the pretrained start, and accuracy stays high along the line between them.
- One network at test time, so inference costs the same as FT. Changing α needs no retraining; the authors recommend $\alpha = 0.5$ when nothing else is known.
- If only the head is fine-tuned, the encoder is shared and the interpolation equals output ensembling of the two models (covered earlier).



Schematic (top-right panel). Wortsman et al.,
[arXiv:2109.01903v3](https://arxiv.org/abs/2109.01903v3), Figure 1.

CLIP ViT-L/14@336px	IN	Mean, 5 shifts
Zero-shot ($\alpha = 0$)	76.6	73.4
Fine-tuned ($\alpha = 1$)	86.2	68.6
WiSE-FT ($\alpha = 0.5$)	86.8	76.9

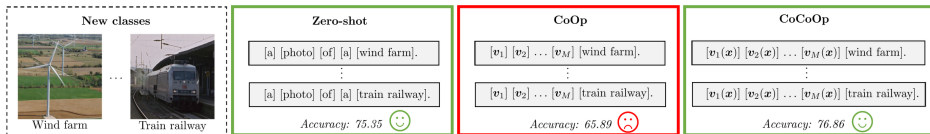
Top-1 (%). Same paper, Table 1.

- ▶ A hyperparameter sweep fine-tunes k models $\theta_1, \dots, \theta_k$ from the same pretrained weights; usually only the best one on held-out data is kept.
- ▶ **Model soup:** average their weights instead. Uniform soup: $\theta_S = \frac{1}{k} \sum_{i=1}^k \theta_i$, where θ_S are the soup's weights. WiSE-FT with $\alpha = 0.5$ is a two-model uniform soup.
- ▶ **Greedy soup**, scored on a held-out split:
 1. sort the models by held-out accuracy, best first;
 2. start the soup with the best model;
 3. add each next model if the soup's held-out accuracy does not drop.
- ▶ Runs from one pretrained start tend to share a low-error basin. Runs with high learning rates can leave it and spoil a uniform soup; the greedy rule rejects them.

CLIP ViT-B/32, $k = 72$	IN	Shifts	Passes
Best single model*	80.38	47.83	1
Uniform soup	79.97	51.45	1
Greedy soup (5 models)	81.03	50.75	1
Output ensemble	81.19	50.77	k

Top-1 (%); Shifts: mean of the five ImageNet shifts; Passes: forward passes per image; * picked by ImageNet accuracy. Wortsman et al., "Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time," ICML 2022, [arXiv:2203.05482v3](https://arxiv.org/abs/2203.05482v3), Table 3.

- ▶ Output ensembling (covered earlier) edges out the greedy soup on ImageNet but needs k forward passes; the soup needs one.
- ▶ **ImageNet:** a greedy soup of 14 of 58 fine-tuned ViT-G/14 models reached 90.94% top-1, above the best single model (90.72%) and the previous state of the art (90.88%).



Prompts and accuracy (%) on unseen classes of SUN397, a scene-recognition dataset. Zhou et al., “Conditional Prompt Learning for Vision-Language Models,” CVPR 2022, [arXiv:2203.05557v2](https://arxiv.org/abs/2203.05557v2), Figure 1(b).

- **CoOp** (Context Optimization; Zhou et al., IJCV 2022) learns the context words of the prompt:

$$t_i = [\mathbf{v}_1][\mathbf{v}_2] \cdots [\mathbf{v}_M][\text{CLASS}_i]$$

$\mathbf{v}_m$: learned vectors of word-embedding size, shared by all classes; $[\text{CLASS}_i]$: name of class i . Only the $\mathbf{v}_m$ are trained, with cross-entropy on a few labelled images; CLIP stays frozen.

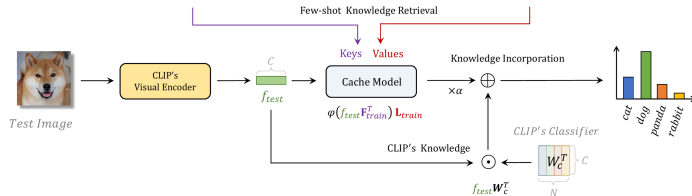
- **CoCoOp** (Conditional Context Optimization): a two-layer network, Meta-Net, maps the image feature to a token π ; each context vector becomes $\mathbf{v}_m(\mathbf{x}) = \mathbf{v}_m + \pi$.

Mean of 11 datasets	Base	New	H
Zero-shot CLIP	69.34	74.22	71.70
CoOp	82.69	63.22	71.66
CoCoOp	80.47	71.69	75.83

ViT-B/16, 16 images per base class. Prompts are learned on half of each dataset’s classes (base) and tested on base and on unseen (new) classes. $H = 2 \cdot \text{Base} \cdot \text{New} / (\text{Base} + \text{New})$, the harmonic mean. Same paper, Table 1(a).

- The prompt depends on the image, so the text encoder runs for every image and class; training is slow (batch size 1). On 7 of the 11 datasets, CoCoOp stays below CLIP on new classes.

Tip-Adapter: A Training-Free Few-Shot Cache



Inference path (lower part of the figure). Zhang et al., "Tip-Adapter: Training-free Adaption of CLIP for Few-shot Classification," ECCV 2022, [arXiv:2207.09519v1](https://arxiv.org/abs/2207.09519v1), Figure 1.

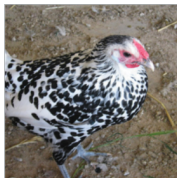
- **Cache** from K labelled images in each of N classes: keys $\mathbf{F}_{\text{train}} \in \mathbb{R}^{NK \times C}$ are their L2-normalised CLIP features, values $\mathbf{L}_{\text{train}} \in \mathbb{R}^{NK \times N}$ their one-hot labels.
- For a test feature $f_{\text{test}} \in \mathbb{R}^{1 \times C}$ (L2-normalised):
- $\mathbf{A} \mathbf{L}_{\text{train}}$: an affinity-weighted vote of the cached labels, a soft k -NN (covered earlier); the same lookup drives retrieval later in this lecture.
- **Tip-Adapter-F** (fine-tuned) trains only the keys, for 20 epochs against CoOp's 200. Top-1 (%), Tables 1 and 5 ([†] as Tip-Adapter reports it):

$$A = \exp(-\beta(1 - f_{\text{test}} \mathbf{F}_{\text{train}}^T))$$

$$\text{logits} = \alpha \mathbf{A} \mathbf{L}_{\text{train}} + f_{\text{test}} \mathbf{W}_c^T$$

$f_{\text{test}} \mathbf{F}_{\text{train}}^T$: cosine similarities to the cached images; β : sharpness; α : weight of the cache term; $\mathbf{W}_c$: CLIP's zero-shot classifier.

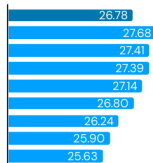
16 shots, ResNet-50	IN	IN-V2	IN-Sketch
Zero-shot CLIP	60.33	53.27	35.44
CoOp [†]	62.95	54.58	31.04
Tip-Adapter	62.03	54.60	35.90
Tip-Adapter-F	65.51	57.11	36.00



Our top prediction: **Hen**
and we say that because...

Average

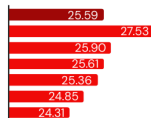
- └ two legs
- └ red, brown, or white feathers
- └ a small body
- └ a small head
- └ two wings
- └ a tail
- └ a beak
- └ a chicken



CLIP's top prediction: **Dalmatian**
but we don't say that because...

Average

- └ black or liver-colored spots
- └ erect ears
- └ long legs
- └ short, stiff hair
- └ a long, tapering tail
- └ a long, slender muzzle



Descriptor scores behind "hen" (left) and the lower scores for "Dalmatian", CLIP's wrong answer (right). Menon and Vondrick, "Visual Classification via Description from Large Language Models," ICLR 2023, [OpenReview](#), Figure 1.

- **Descriptors:** a large language model (LLM), GPT-3 in the paper, is asked "What are useful features for distinguishing a {category name} in a photo?" Its list is the descriptor set $D(c)$ of class c .
- **Score:** CLIP compares the image $\mathbf{x}$ with each text "{category_name} which (is/has/etc) {descriptor}":

$$s(c, \mathbf{x}) = \frac{1}{|D(c)|} \sum_{d \in D(c)} \phi(d, \mathbf{x})$$

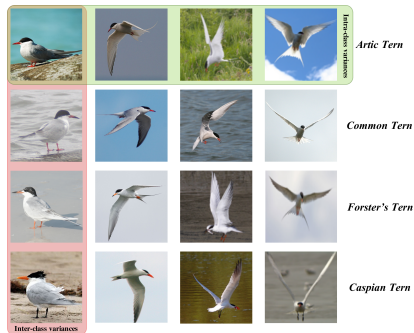
$\phi(d, \mathbf{x})$: CLIP similarity of descriptor d and image $\mathbf{x}$. The predicted class is $\arg \max_c s(c, \mathbf{x})$.

- No training. The descriptor scores explain each decision, and editing $D(c)$ edits the classifier.
- Table 1: ImageNet top-1 from 58.46% to 62.97% (ViT-B/32) and from 72.97% to 76.16% (ViT-L/14@336px); ImageNetV2 from 66.58% to 70.32% (ViT-L/14@336px). The baseline is CLIP with the bare class name.
- **CuPL** (Customized Prompts via Language models; Pratt et al., ICCV 2023) uses LLM-written sentences as prompts: 76.69% on ImageNet vs 75.54% for CLIP's 80 hand-written templates (ViT-L/14, Table 1).

Numbers from the earlier slides of this section (top-1, %; shifts: mean over the shifted test sets).

Recipe	What is trained	Labels	Test-time cost	Evidence (method vs baseline)
FT	all weights	full training set	one network	IN 86.2 vs 76.6, shifts 68.6 vs 73.4 (zero-shot)
LP-FT	head, then all weights	full training set	one network	10-shift OOD 68.9 vs 59.3 (FT)
WiSE-FT	FT, then interpolation with θ_0	full training set	one network	shifts 76.9 vs 68.6, IN 86.8 vs 86.2 (FT)
Greedy soup	k FT runs, averaged	full set, with a held-out split	one network	shifts 50.75 vs 47.83, IN 81.03 vs 80.38 (best single model by ImageNet accuracy)
CoOp	M context vectors	16 per base class	fixed prompts	new classes 63.22 vs 74.22 (CLIP)
CoCoOp	context vectors and Meta-Net	16 per base class	prompts per image	new classes 71.69 vs 74.22, H 75.83 vs 71.70 (CLIP)
Tip-Adapter	nothing	16 per class	two extra matrix products	IN 62.03 vs 60.33 (zero-shot CLIP)
Descriptions	nothing	none	one score per descriptor	IN-V2 70.32 vs 66.58 (class name only)

- ▶ **Fine-grained recognition** assigns an image to one of many subordinate categories of one basic-level category: bird species, car models, aircraft variants.
- ▶ **Small inter-class variation.** Classes share the same basic set of parts. In the Caltech-UCSD Birds dataset (CUB-200-2011), some pairs of species, many sparrows among them, are “nearly visually indistinguishable, even for expert bird watchers.”
- ▶ **Large intra-class variation.** One species appears perched, swimming or flying, occluded, in any light and background.
- ▶ **Labels need experts.** Experts found 494 wrong labels in CUB-200-2011, about 4% of its 11,788 images; citizen-science birders made 2–4 times fewer labelling errors than paid crowd workers.



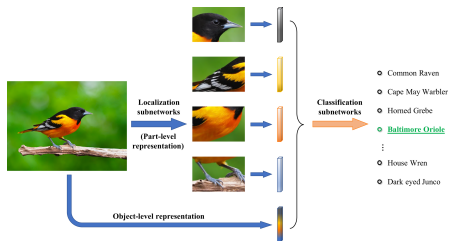
Four tern species from CUB-200-2011, one per row. Wei et al., “Fine-Grained Image Analysis with Deep Learning: A Survey,” TPAMI 2022, [arXiv:2111.06119v2](https://arxiv.org/abs/2111.06119v2), Figure 4.

Wah et al., “The Caltech-UCSD Birds-200-2011 Dataset,” Caltech technical report 2011, Sec. 1; Van Horn et al., “Building a Bird Recognition App and Large Scale Dataset With Citizen Scientists,” CVPR 2015, Sec. 1 and 6.

These benchmarks recur throughout this section; each reports classification accuracy on a fixed test split.

Dataset	Classes	Images	What makes it hard
CUB-200-2011	200 species	11,788	about 60 images per class; look-alike species; 15 part locations and 312 binary attributes annotated per image
Stanford Cars	196 classes	16,185	classes at make, model and year level, such as “2012 Tesla Model S”
FGVC-Aircraft	100 variants	10,000	variants of 70 families from 30 manufacturers; Boeing variants such as -200, -300 and -400 differ mostly in length
NABirds	555 categories	48,562	North American species split by sex, age or plumage into visual categories; expert-curated labels
iNaturalist 2021 (iNat21)	10,000 species	2.7M (train)	the whole tree of life; train and test split by submission date, test from the latest year; a supervised ResNet-50 reaches 76.0%

CUB, Stanford Cars and FGVC-Aircraft return later in this lecture. Wah et al. 2011; Krause et al., “3D Object Representations for Fine-Grained Categorization,” ICCV Workshops 2013, and its dataset page (196 released classes); Maji et al., “Fine-Grained Visual Classification of Aircraft,” 2013, [arXiv:1306.5151v1](#); Van Horn et al., CVPR 2015, Sec. 3–4; Van Horn et al., “Benchmarking Representation Learning for Natural World Image Collections,” CVPR 2021, [arXiv:2103.16483v2](#), Sec. 3.1 and Tables 1–2.



Localisation subnetworks crop parts; part-level and object-level features are classified together. Wei et al., [arXiv:2111.06119v2](#), Figure 8.

- **Localise, then classify.** Early pipelines trained part detectors on annotated parts (part-based R-CNN: 73.9% on CUB, no bounding box at test time). From 2016 the trend was weak supervision: parts learned from image labels alone, often by attention modules that zoom from coarse to fine regions.
- **Bilinear pooling** sums outer products of two CNN feature maps over all image locations l :

$$\Phi(I) = \sum_l \mathbf{a}_l \mathbf{b}_l^T \in \mathbb{R}^{M \times N}, \quad \mathbf{y} = \text{sign}(\mathbf{x}) \sqrt{|\mathbf{x}|}, \quad \mathbf{z} = \mathbf{y} / \|\mathbf{y}\|_2$$

$\mathbf{a}_l \in \mathbb{R}^M$, $\mathbf{b}_l \in \mathbb{R}^N$: features of streams A and B at location l of image I ; $\mathbf{x}$: $\Phi(I)$ flattened to MN entries; sign, square root and $|\cdot|$ act element-wise; a linear classifier reads $\mathbf{z}$.

- Entry (m, n) of Φ correlates channel m of A with channel n of B over the image. With A acting as a part detector and B as a local feature extractor, Φ models part-feature interactions without part labels.
- With image labels only, a fine-tuned bilinear CNN reached **84.1%** on CUB, against 73.9% and 75.7% for part-annotated pipelines built on the older AlexNet.

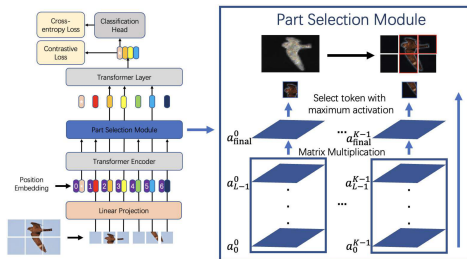
Lin et al., “Bilinear CNN Models for Fine-Grained Visual Recognition,” ICCV 2015, [OpenAccess](#), Sec. 1–3; Zhang et al., “Part-based R-CNNs for Fine-grained Category Detection,” ECCV 2014, [arXiv:1407.3867v1](#); Wei et al., Sec. 5.1.

- **Backbone:** ViT-B/16 (covered earlier) pretrained on ImageNet-21k (about 14M images, 21k classes), fine-tuned at 448 pixels on overlapping patches (size $P = 16$, stride $S = 12$): $37^2 = 1369$ patches instead of $28^2 = 784$, keeping local structure that a non-overlapping split cuts apart.
- **Part Selection Module (PSM):** multiply the raw attention matrices of all layers before the last; for each head, keep the patch with the largest entry in the class-token row:

$$\mathbf{a}_{\text{final}}^k = \mathbf{a}_{L-1}^k \cdots \mathbf{a}_1^k \mathbf{a}_0^k, \quad j_k = \arg \max_{j \geq 1} (\mathbf{a}_{\text{final}}^k)_{0j}$$

$\mathbf{a}_l^k$: attention matrix of head k in layer l (index 0 is the class token); j_k : patch kept for head k . The last layer sees only the class token and the K kept patches.

- **Contrastive loss** on the class tokens, added to cross-entropy: same-class pairs are pulled toward cosine similarity 1, and a different-class pair is pushed apart only while its similarity exceeds a margin $\alpha = 0.4$. Pair losses are treated in detail later in this lecture.
- **CUB:** 91.7% against 90.3% for the plain ViT-B/16 (PSM +0.7, loss +0.5, overlap +0.2 for $2.7\times$ the training time). On Stanford Cars its 94.8% trails two 2020 CNN methods (95.1%, 95.3%).



The figure draws a non-overlapping split. He et al., “TransFG: A Transformer Architecture for Fine-grained Recognition,” AAAI 2022, [arXiv:2103.07976v5](https://arxiv.org/abs/2103.07976v5), Figure 2; results from Tables 1 and 5–7.

Linear probes on frozen features (covered earlier), accuracy % (mean per class for Aircraft). Cars, Aircraft, CUB: logistic regression (Table 8); iNat21: trained linear layer (Table 7).

ViT features	Cars	Aircraft	CUB	iNat21
MAE H/14	56.1	63.4	57.2	32.3
DINO B/8	76.6	70.6	81.7	68.3
iBOT L/16	71.8	72.4	82.1	74.6
OpenCLIP G/14	96.1	80.2	89.9	76.0
DINOv2 g/14	91.4	87.2	91.6	85.7

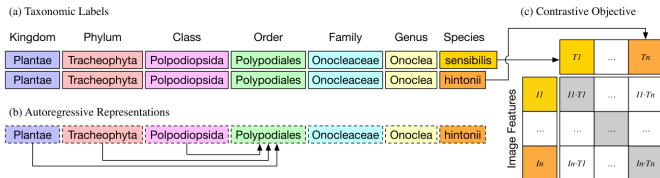
- ▶ DINOv2 leads the best earlier self-supervised features by 14.8 points on both Cars and Aircraft, and beats text-supervised OpenCLIP on Aircraft and CUB.
- ▶ **Caveat:** LVD-142M, DINOv2's curated training set (covered earlier), includes 1M web images retrieved near the training split of each of Cars, Aircraft and CUB. iNaturalist was left out of that retrieval, and there DINOv2 leads OpenCLIP by 9.7 points.

CLIP ViT-L/14@336px, accuracy % (mean per class for Aircraft, Pets, Flowers)

Dataset	Zero-shot	Probe
FGVC-Aircraft	37.2	71.6
Birdsnap	49.5	79.9
Flowers102	78.3	99.2
Stanford Cars	78.8	91.5
Oxford Pets	93.5	95.1

- ▶ The probe reads CLIP's image features before the final projection, and zero-shot CLIP is also a linear classifier on them, so the fine-grained weakness (covered earlier) comes from the class weights: weights from class-name prompts against weights fitted to labelled images. The gap is 34.4 points on Aircraft and 1.6 on Pets.
- ▶ Across CLIP's evaluation datasets, zero-shot accuracy is mostly 10–25 points below the linear probe.

Oquab et al., "DINOv2: Learning Robust Visual Features without Supervision," TMLR 2024, [arXiv:2304.07193v2](#), Tables 7, 8, 15 and 18; Radford et al., "Learning Transferable Visual Models From Natural Language Supervision," ICML 2021, [arXiv:2103.00020v1](#), Tables 10–11 and Figure 8.



Stevens et al., "BioCLIP: A Vision Foundation Model for the Tree of Life," CVPR 2024, [arXiv:2311.18803v3](https://arxiv.org/abs/2311.18803v3), Figure 1(a)–(c); Tables 1, 3 and 5.

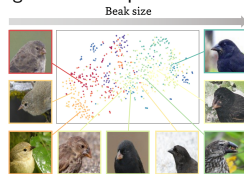
- ▶ **TreeOfLife-10M:** 10.4M images of 454,103 taxa, merged from iNat21 (2.7M), BIOSCAN-1M laboratory photos of insects (1.1M) and the Encyclopedia of Life (6.6M), relabelled into one taxonomy.
- ▶ **Taxonomic name:** the seven ranks from kingdom to species joined into one string. The text encoder is causal (each token attends only to earlier tokens), so each rank is encoded given the ranks above it (panel b), and an unseen species can reuse what was learned for its genus and family.
- ▶ **Training:** the CLIP objective (covered earlier), starting from OpenAI's CLIP ViT-B/16; 100 epochs at batch size 32,768.
- ▶ **Text types:** common (black-billed magpie), scientific (*Pica hudsonia*), taxonomic (Animalia Chordata Aves Passeriformes Corvidae *Pica hudsonia*), and combinations such as taxonomic plus common name.
- ▶ **Mixed text types:** each training step pairs an image with one of its text types drawn at random, so other name types can be used at test time.
- ▶ On 400 at-risk species excluded from training, a model trained on scientific names alone scores 22.3% with scientific-name prompts and 4.5% with taxonomic ones; mixed training scores 24.9–30.9% for every prompt type (1M-image ablation).

Mean top-1 % over 10 biology tasks, all ViT-B/16 (Stevens et al., Table 4)

Model	Zero-shot	1-shot	5-shot
CLIP	21.9	33.6	51.5
OpenCLIP	20.4	36.7	55.9
DINO (image only)	n/a	36.2	55.4
BioCLIP, iNat21 data only	31.7	47.5	65.1
BioCLIP	39.4	50.3	68.8

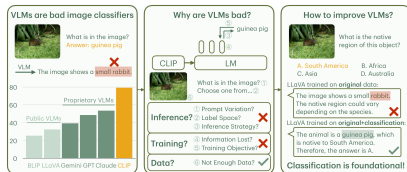
- **k-shot**: each class is the mean of its k image embeddings, and a test image takes the class with the nearest mean. For zero-shot prompts, CLIP and OpenCLIP get common names, their best name type.
- Trained on iNat21 alone (2.7M images, 10,000 species), the same recipe reaches 31.7 zero-shot; the larger and broader TreeOfLife-10M adds the remaining 7.7 points.

- **BioCLIP 2** scales the recipe: TreeOfLife-200M (214M images, 952K taxa), a ViT-L/14 encoder, and 26M replayed LAION-2B image-text pairs.
- Mean zero-shot accuracy over its 10 species tasks rises from 37.6 (BioCLIP) to 55.6.
- With no trait labels, the embeddings order Darwin's finches by beak size (below) and keep life stage and sex orthogonal to the species differences.



Gu et al., "BioCLIP 2: Emergent Properties from Scaling Hierarchical Contrastive Learning," NeurIPS 2025, [arXiv:2505.23883v2](https://arxiv.org/abs/2505.23883v2), Figure 1 (left); Table 1.

- Vision-language models (VLMs, covered earlier) such as LLaVA-1.5 feed image-encoder outputs through a small projector into a language model; LLaVA-1.5 uses CLIP-L (CLIP ViT-L/14 at 336 pixels). With all class names in the prompt, an answer counts as correct if it contains the true label.
- **Gap:** on Stanford Cars every VLM tested trails CLIP-L used zero-shot. LLaVA-1.5-7B scores 10.2% on Flowers102, where its own encoder scores 76.0%.
- **The information is kept:** a linear probe on LLaVA-1.5-7B's last-layer features reaches 82.8% on Cars, and training only the projector to generate the label text reaches 90.4% (CLIP-L probe: 91.5%).
- **Data decides:** ImageNet classes named fewer than 10 times in LLaVA's training data score 3.0%, classes named over 10,000 times 82.7% (rank correlation 0.76).
- **Fix:** adding ImageNet's 1.28M labelled images to LLaVA's 665K instruction-tuning examples lifts ImageNet accuracy from 22.8% to 84.4% and ImageWikiQA (questions that need the object's class) from 38.0% to 49.8%. ImageNet data alone drops ImageWikiQA to 30.6%.



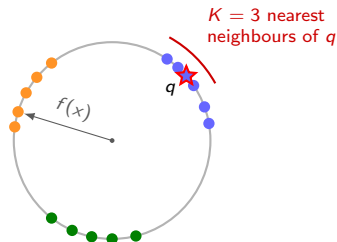
Zhang et al., "Why are Visually-Grounded Language Models Bad at Image Classification?" NeurIPS 2024, [arXiv:2405.18415v2](https://arxiv.org/abs/2405.18415v2), Figure 1; Tables 1, 3, 5 and Sec. 3.3.

Accuracy (%) with all class names in the prompt (Table 1)

Model	ImageNet	Flowers102	Cars
LLaVA-1.5-7B	n/a	10.2	0.0
Claude 3 Opus	51.1	58.3	45.1
Gemini Pro Vision	56.0	62.0	66.6
GPT-4 Turbo	60.6	79.9	58.2
CLIP-L, zero-shot	74.8	76.0	77.5

n/a: 1,000 class names do not fit LLaVA's 4K-token context.

- **Goal:** learn an embedding $f : \mathcal{X} \rightarrow \mathbb{R}^d$ with $\|f(x)\|_2 = 1$, so that the distance $d(x, x') = \|f(x) - f(x')\|_2$ is small for two images of the same class and large otherwise. For unit vectors $d^2 = 2 - 2 \cos(f(x), f(x'))$, so distance and cosine similarity rank neighbours identically (covered earlier).
- **Why not a softmax head:** it has one output per training class and cannot name a class it never saw. Retrieval, face verification (same person or not) and re-identification (later in this lecture) meet new classes at test time, so they compare embeddings by k -NN search (covered earlier) or with a distance threshold.
- **Zero-shot split:** train on the first half of the classes, test on the disjoint second half. CUB-200-2011 (earlier in this lecture): 100 + 100 species; Cars196 (Stanford Cars): 98 + 98 models; Stanford Online Products (SOP): 120,053 eBay images of 11,318 + 11,316 products.



Unit-norm embeddings lie on a sphere; colours are classes. The query q (star, red outline) retrieves its $K = 3$ nearest neighbours, all of its own class.

Split: Song et al., “Deep Metric Learning via Lifted Structured Feature Embedding,” CVPR 2016, [OpenAccess](#), Sec. 7.

- **Recall@K**: each test image queries the rest of the test set and scores 1 if an image of its class is among its K nearest neighbours:

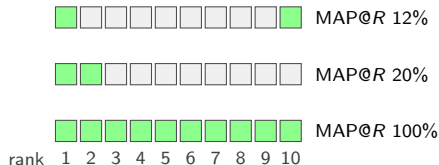
$$\text{Recall@K} = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}[\exists j \in \mathcal{N}_K(q) : y_j = y_q]$$

Q : the test queries; $\mathcal{N}_K(q)$: the K test images nearest to q , excluding q ; y : class label; $\mathbf{1}[\cdot]$: 1 if the condition holds, else 0.

- **MAP@R** (mean average precision at R): for a query with R same-class images in the test set, look at its R nearest neighbours:

$$\text{MAP@R} = \frac{1}{R} \sum_{i=1}^R P(i)$$

$P(i)$: the precision at rank i if result i is correct, else 0; averaged over queries.



One query with $R = 10$ same-class images; green: correct result. Recall@1 is 100% in all three rows, yet MAP@R separates them: $(1 + 2/10)/10 = 12\%$ in the top row. Recall@K: Song et al., CVPR 2016, Sec. 6. After Musgrave et al., "A Metric Learning Reality Check," ECCV 2020, [arXiv:2003.08505v3](https://arxiv.org/abs/2003.08505v3), Table 3.

- Musgrave et al. report MAP@R because it rewards well-clustered embeddings and is steadier during training: lag-one autocorrelation of validation accuracy 0.81, against 0.73 for Recall@1 (Sec. 3.2).

- **Contrastive loss** (Hadsell et al., CVPR 2006), a margin-based pair loss distinct from InfoNCE and NT-Xent (covered earlier), on a pair (x_i, x_j) with $y_{ij} = 1$ if both images share a class, else 0:

$$\mathcal{L}_{\text{con}} = y_{ij} \frac{1}{2} d_{ij}^2 + (1 - y_{ij}) \frac{1}{2} [m - d_{ij}]_+^2$$

$d_{ij} = d(x_i, x_j)$; $[u]_+ = \max(0, u)$; the **margin** $m > 0$ is the gap the loss demands: a negative pair farther apart than m adds no loss.

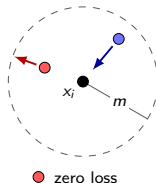
- **Triplet loss** (Schroff et al., FaceNet, CVPR 2015) on an anchor a , a positive p (same class) and a negative n (another class):

$$\mathcal{L}_{\text{tri}} = [d(a, p)^2 - d(a, n)^2 + m]_+$$

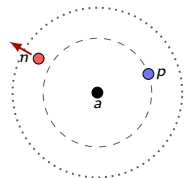
summed over triplets. FaceNet used $m = 0.2$ on 128-d unit-norm face embeddings.

- The triplet loss constrains only relative distances, so a class may spread out as long as other classes stay farther away; the contrastive loss applies one radius to every pair.

contrastive: one radius m



triplet: radius from $d(a, p)$



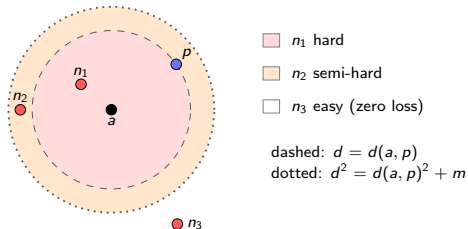
Black: anchor; blue: same class; red: other class. Left: same-class points are pulled in and other classes pushed out to radius m . Right: the triplet margin is measured from the positive's distance.

Hadsell et al., "Dimensionality Reduction by Learning an Invariant Mapping," CVPR 2006, [doi:10.1109/CVPR.2006.100](https://doi.org/10.1109/CVPR.2006.100), Eq. 4 (written here with $y_{ij} = 1$ for a similar pair); Schroff et al., "FaceNet: A Unified Embedding for Face Recognition and Clustering," CVPR 2015, [arXiv:1503.03832v3](https://arxiv.org/abs/1503.03832v3), Eq. 3.

- ▶ **Most triplets teach nothing:** their number grows cubically with the dataset, and the network soon satisfies the margin for a large fraction of them. Those give zero loss and no gradient, but still cost a forward pass.
- ▶ **The hardest negatives mislead:** the closest negative is often a mislabelled or outlier image, and always picking it early in training can collapse the embedding.
- ▶ **Semi-hard mining (FaceNet):** keep negatives with $d(a, p)^2 < d(a, n)^2 < d(a, p)^2 + m$, found online inside batches of about 1,800 images (about 40 per identity).
- ▶ **Batch hard (Hermans et al., 2017):** sample P classes with S images each; every anchor uses its hardest positive and hardest negative in the batch:

$$\mathcal{L}_{\text{BH}} = \sum_a \left[m + \max_{p: y_p = y_a} d(a, p) - \min_{n: y_n \neq y_a} d(a, n) \right]_+$$

with a running over all PS images. On a MARS validation split (video-based person re-identification, covered later in this lecture), $m = 0.2$: random triplets 41.7 mAP, batch hard **65.3** mAP.



Negatives seen from the anchor: closer than the positive (hard), inside the margin (semi-hard), or beyond it with zero loss (easy).

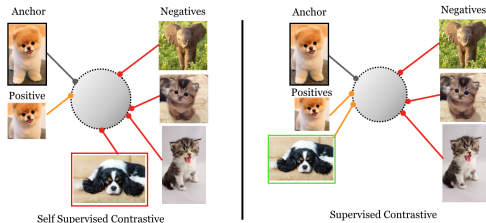
Schroff et al., [arXiv:1503.03832v3](#), Sec. 3.2; Hermans et al., "In Defense of the Triplet Loss for Person Re-Identification," arXiv 2017, [arXiv:1703.07737v4](#), Sec. 2, Eq. 5, Table 1.

- ▶ NT-Xent (covered earlier) gives each anchor one positive, its other augmented view. **SupCon** (Khosla et al., NeurIPS 2020) also treats every other batch image of the anchor's class as a positive:

$$\mathcal{L}_{\text{out}}^{\text{sup}} = \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(z_i \cdot z_p / \tau)}{\sum_{a \in A(i)} \exp(z_i \cdot z_a / \tau)}$$

I : the $2N$ augmented views of N images;
 $A(i) = I \setminus \{i\}$; $P(i) \subseteq A(i)$: the views with the label of i ; z : unit-norm projection-head output; τ : temperature.

- ▶ **Where the average sits matters:** putting the sum over $P(i)$ and the $1/|P(i)|$ inside the log gives 67.4% ImageNet top-1, against 78.7% for the form above (ResNet-50, batch 6,144).
- ▶ Many positives and negatives per anchor weight hard examples through the gradient, so no explicit mining is needed.
- ▶ Two stages: SupCon trains the encoder, then a linear probe (covered earlier) is fit on the frozen features.



Khosla et al., "Supervised Contrastive Learning," NeurIPS 2020, [arXiv:2004.11362v5](https://arxiv.org/abs/2004.11362v5), Figure 2; numbers from Sec. 3.2 and Tables 1 and 3.

ImageNet top-1 (%)	Cross-entropy	SupCon
ResNet-50	78.2	78.7
ResNet-200	80.9	81.4

Same augmentation within each row (Table 3).

- ▶ **Proxy**: a learnable vector p_c for each class c , trained with the network; each image is compared with the C proxies instead of with other images. With M training images an epoch costs $O(MC)$ comparisons, against $O(M^2)$ for pairs and $O(M^3)$ for triplets.
- ▶ **Proxy-NCA** (Movshovitz-Attias et al., ICCV 2017; NCA: neighbourhood components analysis):

$$\ell(x) = -\log \frac{e^{s(x, p^+)}}{\sum_{p^- \in P^-} e^{s(x, p^-)}}$$

s : cosine similarity; p^+ : the proxy of the class of x ; P^- : the other proxies. It converged about three times as fast as triplet-based losses.

- ▶ **Proxy Anchor** (Kim et al., CVPR 2020) makes each proxy the anchor and compares it with every embedding in the batch:

$$\ell(X) = \frac{1}{|P^+|} \sum_{p \in P^+} \log \left(1 + \sum_{x \in X_p^+} e^{-\alpha(s(x, p) - \delta)} \right) + \frac{1}{|P|} \sum_{p \in P} \log \left(1 + \sum_{x \in X_p^-} e^{\alpha(s(x, p) + \delta)} \right)$$

X_p^+ (X_p^-): batch embeddings of (not of) the class of p ; P^+ : proxies of the classes in the batch; P : all proxies; scale $\alpha = 32$, margin $\delta = 0.1$. The log-sum-exp gives harder examples larger gradients.

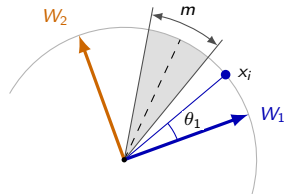
Movshovitz-Attias et al., "No Fuss Distance Metric Learning using Proxies," ICCV 2017, [arXiv:1703.07464v3](#); Kim et al., "Proxy Anchor Loss for Deep Metric Learning," CVPR 2020, [arXiv:2003.13911v1](#), Eqs. 1 and 4.

- **Classification as proxy learning:** drop the bias and normalise the class weights W_j and the feature x_i ; the logit becomes $\cos \theta_j$, and each W_j acts as a proxy for class j (Zhai and Wu, BMVC 2019). At test time x_i is the embedding.
- **ArcFace** (Deng et al., CVPR 2019) adds a margin m to the angle of the true class:

$$\mathcal{L}_{\text{Arc}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s \cos(\theta_{y_i} + m)}}{e^{s \cos(\theta_{y_i} + m)} + \sum_{j \neq y_i} e^{s \cos \theta_j}}$$

N : batch size; y_i : class of image i ; s : a scale, since cosines in $[-1, 1]$ cannot make the softmax confident. ArcFace used $s = 64$, $m = 0.5$, 512-d features.

- **Target logit:** normalised softmax $\cos \theta_y$; CosFace $\cos \theta_y - m$ (Wang et al., CVPR 2018); ArcFace $\cos(\theta_y + m)$, a margin in angle, i.e. geodesic distance on the sphere.
- **No pair sampling:** its authors name the “combinatorial explosion” of triplets as a drawback of the triplet loss.



Two classes on the unit circle during training.
Dashed: the boundary without a margin (the bisector). ArcFace makes each class win by angle m , so the two boundaries open a band of width m (grey).

Deng et al., “ArcFace: Additive Angular Margin Loss for Deep Face Recognition,” CVPR 2019, [OpenAccess](#), Eq. 3; Zhai and Wu, “Classification is a Strong Baseline for Deep Metric Learning,” BMVC 2019, [arXiv:1811.12649v2](#); Wang et al., “CosFace: Large Margin Cosine Loss for Deep Face Recognition,” CVPR 2018, [arXiv:1801.09414v2](#).

- ▶ Musgrave et al. (ECCV 2020) re-ran 13 losses from 2006–2019 under one protocol: BN-Inception (Inception with batch normalisation) pretrained on ImageNet, 128-d embeddings, batch 32, one augmentation recipe, hyperparameters from Bayesian optimisation with 4-fold cross-validation on the training classes, 10 runs per loss.
- ▶ **What papers had done:** changed backbone, embedding size or augmentation between compared methods; tuned on the test set; cited weak baselines, such as a triplet margin of 1 where about 0.1 works best.
- ▶ **Hidden details:** one paper's code froze the backbone's BatchNorm parameters, worth 2 points on CUB according to its authors, without saying so in the paper.
- ▶ **Finding:** newer losses “perform marginally better than, and sometimes on par with, classic methods” (table).

MAP@R (%), mean of 10 runs, 512-d (four 128-d models concatenated)

Loss (year)	CUB	Cars	SOP
No training (PCA to 512-d)	14.21	5.91	23.44
Contrastive pair (2006)	26.53	24.89	44.39
Triplet (2006)	23.69	23.02	43.37
NT-Xent (2016)	25.09	24.40	45.31
Proxy-NCA (2017)	24.21	25.38	47.22
Norm. softmax (2017)	25.25	26.00	47.13
CosFace (2018)	26.70	27.57	46.92
ArcFace (2019)	26.45	27.22	47.41

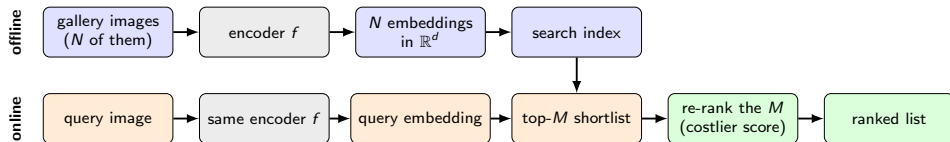
Musgrave et al., [arXiv:2003.08505v3](#), Tables 4–7, Secs. 2–5.

Recall@1 (%) on the zero-shot splits, each paper's own protocol. ViT-S/16, ViT-B/16: small and base ViTs with 16×16 patches.

Method	Idea	Backbone, pretraining	Dim	CUB	Cars	SOP
Proxy Anchor (2020)	proxy loss (earlier)	BN-Inception, ImageNet-1k	512	68.4	86.1	79.1
No training	frozen features	ViT-S/16, ImageNet-21k	384	83.1	47.8	62.1
Hyp-ViT (2022)	hyperbolic embeddings	ViT-S/16, ImageNet-21k	384	85.6	86.5	85.9
Hyp-DINO (2022)	same, from DINO weights	ViT-S/16, DINO on ImageNet-1k	384	80.9	89.2	85.1
RS@k (2022)	smoothed Recall@k as the loss	ViT-B/16, ImageNet-21k	512	n/a	89.5	88.0
Unicom (2023)	ArcFace on 1M pseudo-classes	ViT-B/16, LAION-400M	768	88.8	97.7	88.8

- **Hyp-ViT**: embeddings in a Poincaré ball, a hyperbolic space whose volume grows exponentially with radius (suited to tree-like hierarchies), trained with an NT-Xent-style loss on hyperbolic distances. **RS@k** (Recall@k surrogate): sigmoids replace the step functions that count ranks; batches of 4,000 images (392 on Cars196); SiMix mixes the similarities of same-class pairs. **Unicom**: CLIP features (covered earlier) cluster LAION-400M into one million pseudo-classes.
- **On CUB, the pretraining data did most of the work**: frozen ImageNet-21k ViT-S features beat every CNN method in Ermolov et al.'s table on CUB (best 69.0); the authors suspect ImageNet-21k's bird classes. On Cars and SOP frozen features lag far behind, so training still matters.

Kim et al., [arXiv:2003.13911v1](#), Tables 2–3; Ermolov et al., “Hyperbolic Vision Transformers: Combining Improvements in Metric Learning,” CVPR 2022, [arXiv:2203.10833v2](#), Table 3; Patel et al., “Recall@k Surrogate Loss with Large Batches and Similarity Mixup,” CVPR 2022, [arXiv:2108.11179v2](#), Table 2; An et al., “Unicom: Universal and Compact Representation Learning for Image Retrieval,” ICLR 2023, [arXiv:2304.05884v1](#), Tables 4 and 7.



Offline, once per gallery image: embed and index. Online, per query: embed the query, search, and re-rank only the shortlist.

- ▶ **Query and gallery:** the query is the image we search with; the gallery (also called the database) is the collection we rank. Both pass through the same embedding f (earlier in this lecture), and similarity is the cosine of two L2-normalised embeddings.
- ▶ **Instance-level** retrieval looks for the same object or place under a new viewpoint, lighting or occlusion; **category-level** retrieval looks for other members of a class. Landmark search and re-identification (later in this lecture) are instance-level.
- ▶ **Two stages:** a **global descriptor** (one vector per image) ranks the whole gallery; a costlier score then re-ranks only the top M (the systems in this section use $M = 400$ to 1,000).
- ▶ **Scoring:** the AP of each query's ranked list (covered earlier), averaged over queries, gives mAP; mAP@ k computes AP on the top k results only.

- ▶ **ROxford and RParis** (Revisited Oxford and Paris, 2018): 70 queries each, over 4,993 and 6,322 gallery images. For each query, every gallery image is labelled Easy, Hard, Unclear or Negative.
- ▶ **Protocols:** the positives (images that should be retrieved) are the Easy images in *Easy*, Easy and Hard in *Medium*, and Hard only in *Hard*, which ignores Easy ones; Unclear images are always ignored. Easy is nearly saturated, so results report Medium (M) and Hard (H).
- ▶ **R1M:** 1,001,001 distractors (images that match no query), the most confusing million of about 4.1M Flickr photos; they drop a fine-tuned ResNet-101 descriptor from 64.7 to 45.2 mAP on ROxford Medium.
- ▶ **GLDv2** (Google Landmarks Dataset v2, 2020): 4.1M training images of 203k landmarks, a 762k-image gallery, and 118k queries of which only 1.1% show a landmark; scored with mAP@100.

Weyand et al., “Google Landmarks Dataset v2 – A Large-Scale Benchmark for Instance-Level Recognition and Retrieval,” CVPR 2020, [arXiv:2004.01804v2](#), Section 3.



Queries (blue) with Easy (dark green), Hard (light green) and Unclear (yellow) gallery images. Radenović et al., “Revisiting Oxford and Paris: Large-Scale Image Retrieval Benchmarking,” CVPR 2018, [arXiv:1803.11285v1](#), Figure 3; counts from Section 2, mAP from Table 5.

Train-test overlap (2024). GLDv2-clean (a cleaned 1.58M-image subset of GLDv2’s training images, the most popular training set for landmark retrieval) contains the landmarks of 38 of the 70 ROxford queries and 36 of the 70 RParis ones. Retrained by Song et al. without those 18 landmarks (1,565 images), five methods fall 10.6 to 29.7 mAP below their published ROxford Hard results. Song et al., “On Train-Test Class Overlap and Detection for Image Retrieval,” CVPR 2024, [arXiv:2404.01524v1](#), Section 3 and Table 4.

- ▶ A global descriptor pools the last convolutional feature map into one vector. **GeM** (generalised-mean) pooling computes, for each channel k ,

$$f_k = \left(\frac{1}{|\mathcal{X}_k|} \sum_{x \in \mathcal{X}_k} x^{p_k} \right)^{1/p_k}, \quad k = 1, \dots, K,$$

where $\mathcal{X}_k$ holds the $W \times H$ non-negative (post-ReLU) activations of channel k , K is the number of channels (the descriptor length) and p_k is the pooling exponent.

- ▶ $p_k = 1$ gives average pooling and $p_k \rightarrow \infty$ gives max pooling. GeM learns p by backpropagation, and one p shared by all channels worked better than one per channel. A larger p concentrates the response on fewer, more localised regions (right). The vector $(f_1, \dots, f_K)$ is L2-normalised and compared by inner product.
- ▶ **Whitening**: a linear map after pooling that decorrelates and rescales the dimensions. Learned from matching pairs, it lifts fine-tuned VGG with GeM on Oxford5k from 82.0 to 85.9 mAP; PCA whitening (PCA covered earlier) reaches 83.1.
- ▶ **Update (2023)**: trained with ArcFace (earlier in this lecture), a learnable p drifts below the test-time optimum of 4.0, so SuperGlobal's GeM+ (Shao et al., ICCV 2023; later in this lecture) re-tunes p after training.



$p = 1$

$p = 3$

$p = 10$

Activations x^p of one channel (off-the-shelf VGG) for three values of p . Radenović et al., "Fine-tuning CNN Image Retrieval with No Human Annotation," IEEE TPAMI 2019, [arXiv:1711.02512v2](https://arxiv.org/abs/1711.02512v2), Figure 4.

Fine-tuned AlexNet with learned whitening, original Oxford5k and Paris6k annotations (mAP):

Pooling	Oxford5k	Paris6k
max ($p = \infty$)	62.2	68.9
average ($p = 1$)	61.2	70.8
GeM, p learned ($3 \rightarrow 2.32$)	67.7	75.5

Same paper, Table 1 (whitening numbers: Table 2).

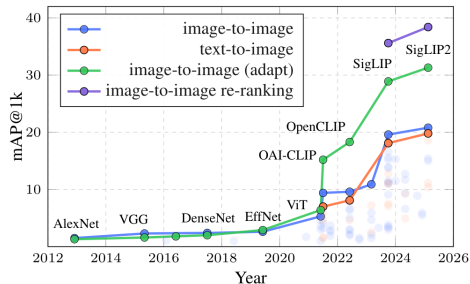
Frozen features, gallery ranked by cosine similarity (mAP):

Model	ROxford		RParis		AmsterTime
	M	H	M	H	
OpenCLIP ViT-G/14	50.7	19.7	79.2	60.2	24.6
MAE ViT-H/14	11.7	2.2	19.9	4.7	4.2
DINO ViT-B/8	40.1	13.7	65.3	35.3	24.6
iBOT ViT-L/16	39.0	12.7	70.7	47.0	26.7
DINOv2 ViT-S/14	68.8	43.2	84.6	68.5	43.5
DINOv2 ViT-B/14	72.9	49.5	90.3	78.6	45.6
DINOv2 ViT-L/14	75.1	54.0	92.7	83.5	50.0
DINOv2 ViT-g/14	73.6	52.3	92.1	82.6	46.7

Oquab et al., “DINOv2: Learning Robust Visual Features without Supervision,” TMLR 2024, [arXiv:2304.07193v2](https://arxiv.org/abs/2304.07193v2), Table 9 (Met columns omitted); data sources from Table 15.

- **Protocol:** no training on the benchmark; one frozen global feature per image and the gallery ranked by cosine similarity. M and H are the Medium and Hard protocols; AmsterTime matches street-view photos of Amsterdam to archival photos of the same places.
- On ROxford Hard, DINOv2 ViT-L/14 reaches 54.0 mAP, against 13.7 for the best earlier self-supervised feature (DINO) and 19.7 for OpenCLIP. Even ViT-S/14 beats OpenCLIP ViT-G/14 in every column. Masked-image-modelling features (MAE, covered earlier) score 2.2.
- **Caveat:** LVD-142M (covered earlier) contains GLDv2-clean plus 6.3M web images retrieved around it, and 1M web images retrieved around each of the ROxford and RParis galleries. Near-duplicates of test images were removed, yet the training data includes GLDv2-clean with its landmarks that overlap ROxford and RParis queries (earlier in this lecture).
- **Implication:** frozen DINOv2 features with cosine search beat every earlier off-the-shelf feature here, but these benchmarks sit inside DINOv2’s training distribution. ILIAS (later in this lecture) tests outside it.

- ▶ A **universal embedding** serves every domain with one model and one index, instead of one specialist model per domain. **UnED** (Universal Embedding Dataset, 2023) merges 8 domains (food, cars, online products, clothing, natural world, artworks, landmarks, retail products) into 349k classes: 2.8M training, 1.4M gallery and 241k query images. Each query is searched against the merged gallery of all domains.
- ▶ Mean Recall@1 over the 8 domains: frozen DINOv2 (768-d) 58.2; one universal CLIP ViT-B/16 fine-tuned to 64-d, 65.9; eight 64-d specialists, each told the query's domain, 70.4. Artworks lag most (universal 12.7, specialist 32.9).
- ▶ **ILIAS** (Instance-Level Image retrieval At Scale, 2025): 1,000 objects with 1,232 queries and 4,715 positives, hidden among 100M Flickr photos collected in 2014. Every object first appeared after 2014, so no distractor can be an unlabelled positive. Metric: mAP@1k.
- ▶ Frozen DINOv2 ViT-L scores 15.3 mAP@1k and SigLIP ViT-L 19.6 (**SigLIP**: a CLIP-style image-text model trained with a pairwise sigmoid loss in place of the batch softmax). One linear layer trained on 1M UnED images lifts SigLIP to 28.9 and leaves DINOv2 at 15.3. Models fine-tuned only on landmarks or products fail on ILIAS.



Best mAP@1k on ILIAS by model release year; “adapt” adds the linear layer trained on UnED. Kordopatis-Zilos et al., “ILIAS: Instance-Level Image retrieval At Scale,” CVPR 2025, [arXiv:2502.11748v3](#), Figure 1; numbers from Sections 3–4 and Table 2.

UnED: Ypsilantis et al., “Towards Universal Image Embeddings: A Large-Scale Dataset and Challenge for Generic Image Representations,” ICCV 2023, [arXiv:2309.01858v1](#), Table 4. SigLIP: Zhai et al., “Sigmoid Loss for Language Image Pre-Training,” ICCV 2023, [arXiv:2303.15343v4](#).

- **Query expansion (QE):** search once, build a new query from the query and its top- n results, and search again:

$$\mathbf{q}' \propto \mathbf{q} + \sum_{i=1}^n (\mathbf{q}^T \mathbf{d}_i)^\alpha \mathbf{d}_i$$

$\mathbf{q}$ is the query descriptor, $\mathbf{d}_i$ the descriptor of the i -th ranked gallery image, $\alpha \geq 0$ sets how strongly the closest neighbours dominate, and $\mathbf{q}'$ is L2-normalised again. $\alpha = 0$ is average QE (**AQE**; Chum et al., ICCV 2007); $\alpha > 0$ is α **QE** (Radenović et al., TPAMI 2019, with $\alpha = 3$, $n = 50$).

- On the fine-tuned ResNet-101 GeM descriptor, α QE raises Medium mAP from 64.7 to 67.2 on ROxford and from 77.2 to 80.7 on RParis.
- QE improves queries that already have positives near the top and harms many low-scoring ones by averaging in unrelated images. On ILIAS, α QE with $\alpha = 1$ lifts SigLIP from 19.6 to 22.1 mAP@1k when few neighbours are aggregated, and drops it to 14.3 when more are.

Revisiting Oxford and Paris, Table 5; GeM paper, Sections 3.5 and 5.3; ILIAS, Sections 5.2–5.3 and Table 4.

- **SuperGlobal** (Shao et al., ICCV 2023) re-ranks the top $M = 400$ with global descriptors only. Each candidate is replaced by a similarity-weighted average with its $K = 9$ nearest neighbours, the query is expanded by max-pooling its refined neighbours, and the final score averages the two similarities. The cost is $O(M^2)$ on vectors already in memory.

ResNet-101	mAP	time	memory
CVNet (local)	53.7 → 67.4	24 s	27.02 GB
SuperGlobal	61.9 → 71.1	0.37 ms	0.04 GB

ROxford+R1M Hard, before → after re-ranking the top 400; re-ranking time per query; feature memory for the ROxford gallery. Shao et al., “Global Features are All You Need for Image Retrieval and Reranking,” ICCV 2023, [arXiv:2308.06954v2](https://arxiv.org/abs/2308.06954v2), Tables 1–2.

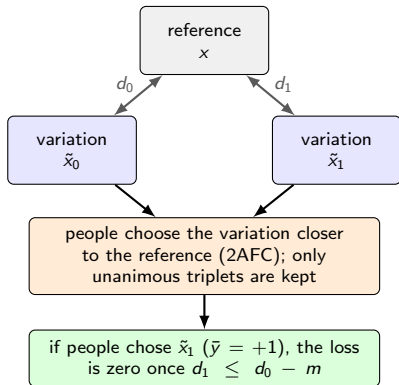
- CVNet (Lee et al., CVPR 2022) scores each pair with 4D convolutions over dense feature correlations.
- Local descriptors (many keypoint-level features per image) still pay off under clutter: AMES (Asymmetric and Memory-Efficient Similarity; Suma et al., ECCV 2024), a transformer that compares 600 query and 100 gallery binary local descriptors, lifts SigLIP on ILIAS from 19.6 to 26.4 mAP@1k (ILIAS, Section 4 and Table 4).

- ▶ LPIPS (covered earlier) was fitted to human judgments of low-level distortions such as blur and noise. Retrieval and generation also need mid-level similarity: pose, layout, shape, colour, number of objects.
- ▶ **NIGHTS** (Novel Image Generations with Human-Tested Similarity): the text-to-image model Stable Diffusion draws a reference and two variations from one prompt, and annotators make a two-alternative forced choice (**2AFC**): which variation is closer to the reference? Keeping only unanimous votes leaves 20,019 triplets.
- ▶ **DreamSim** concatenates DINO, CLIP and OpenCLIP embeddings and fine-tunes them with LoRA (covered earlier) on the 2AFC labels. With $d_j = 1 - \cos(f(x), f(\tilde{x}_j))$:

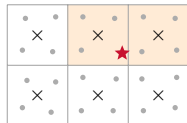
$$\mathcal{L} = \max(0, m - \bar{y}(d_0 - d_1))$$

x : reference; $\tilde{x}_0, \tilde{x}_1$: variations; $\bar{y} = +1$ if people chose $\tilde{x}_1$, else -1 ; $m = 0.05$: the margin of this triplet-style loss.

- ▶ Agreement with people on held-out triplets (chance 50%): LPIPS 70.7%, DINO ViT-B/16 90.1%, untuned ensemble 90.8%, DreamSim 96.2%.

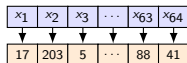


The training signal of DreamSim, drawn for this slide after Fu et al., "DreamSim: Learning New Dimensions of Human Visual Similarity using Synthetic Data," NeurIPS 2023, [arXiv:2306.09344v3](https://arxiv.org/abs/2306.09344v3), Secs. 3–4; numbers from Secs. 3.2 and 4.2 and supplementary Table 5.



k -means centroids ($\times$) define K lists (here 6)

query ($\star$): scan only the $P = 2$ nearest lists



PQ: $d = 768$ cut into 64 sub-vectors of 12 dims, each kept as the index of its nearest of 256 centroids (1 byte): 64 B per vector

Top: an inverted file picks which lists to scan. Bottom: product quantisation stores each scanned vector in 64 bytes.

Jégou et al., "Product Quantization for Nearest Neighbor Search," IEEE TPAMI 2011, doi:10.1109/TPAMI.2010.57; Malkov and Yashunin, "Efficient and Robust Approximate Nearest Neighbor Search Using Hierarchical Navigable Small World Graphs," IEEE TPAMI 2020, doi:10.1109/TPAMI.2018.2889473; Douze et al., "The Faiss library," arXiv 2024 (v4, 2025), arXiv:2401.08281v4, Sections 5.1 and 5.3.

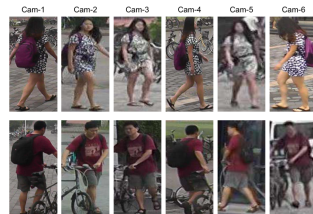
- **Exact search** costs $O(Nd)$ per query. Worked example with $N = 10^9$ gallery vectors of $d = 768$ float32 values: $10^9 \times 768 \times 4\text{B} = 3.07\text{ TB}$ to scan and 7.7×10^{11} multiply-adds for every query.
- **PQ** (product quantisation; Jégou et al., TPAMI 2011) stores each of $m = 64$ sub-vectors as the index of its nearest of 256 k -means centroids (covered earlier): 64 B per vector, so 64 GB of codes for 10^9 vectors, $48\times$ less. **ADC** (asymmetric distance computation) keeps the query exact: one 64×256 table of sub-distances per query, then 64 look-ups and additions per gallery code.
- **IVF** (inverted file): k -means splits the gallery into K lists, one per centroid, and a query scans only its P nearest lists, $K + PN/K$ distances in all. With $K = 65,536$ and $P = 64$: $65,536 + 976,563 \approx 1.04 \times 10^6$ instead of 10^9 . In IVFADC the lists hold PQ codes of each vector's residual to its centroid.
- **HNSW** (hierarchical navigable small world; Malkov and Yashunin, TPAMI 2020) stacks proximity graphs (each vector linked to near neighbours) over nested subsets of the gallery and searches greedily from the sparse top layer down, with logarithmic scaling. **Faiss** (Douze et al., 2024) implements these indexes and advises graphs when memory is no constraint (typically below 1M vectors) and IVF when vectors must be compressed to fit in memory.

- **Task.** Person re-identification (re-ID) ranks a gallery of person crops from other cameras so that crops of the query's identity come first. It is retrieval with a learned embedding (query, gallery and mAP as defined earlier in this lecture).
- **Unseen identities.** Training and test identities are disjoint (MSMT17: 1,041 and 3,060), so the embedding must separate people never seen in training. Deciding whether a query matches anyone at all is an open-set problem; open-set recognition is covered later in this lecture.
- **Cross-camera rule.** In the Market-1501 protocol, gallery crops of the query's identity from the query's own camera are "junk" and ignored, so every counted match crosses cameras.
- **Cumulative matching characteristic (CMC):** the fraction of queries whose first correct match is in the top k ,

$$\text{CMC}(k) = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}[r_q \leq k]$$

Q : query set; r_q : rank of the first correct gallery crop for q ; $\mathbf{1}[\cdot]$: 1 if true, else 0. Rank-1 accuracy is $\text{CMC}(1)$.

- **CMC and mAP.** CMC stops at the first hit, yet a Market-1501 query has 14.8 cross-camera ground truths on average, so papers report mAP next to rank-1.



Two identities, one query per camera. Zheng et al., "Scalable Person Re-Identification: A Benchmark," ICCV 2015, [OpenAccess](#), Figure 4.

Dataset	Identities	Boxes	Cameras
Market-1501	1,501	32,668	6
MSMT17	4,101	126,441	15

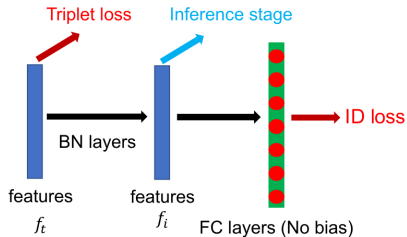
MSMT17: multi-scene, multi-time. Counts: Zheng et al., Section 3.1; Wei et al., "Person Transfer GAN to Bridge Domain Gap for Person Re-Identification," CVPR 2018, [arXiv:1711.08565v2](#), Sections 3.2–3.3.

- **Identity (ID) loss.** Luo et al. train ResNet-50 (covered earlier) as an N -way classifier over the training identities with softmax cross-entropy (covered earlier) and **label smoothing**:

$$q_i = \begin{cases} 1 - \frac{N-1}{N} \varepsilon & i = y \\ \varepsilon / N & i \neq y \end{cases} \quad \mathcal{L}_{\text{ID}} = - \sum_{i=1}^N q_i \log p_i$$

q_i : target; y : true identity; p_i : predicted probability of identity i ; $\varepsilon = 0.1$. Smoothing curbs overfitting to training identities, which never reappear at test time.

- **Triplet loss** with batch-hard mining (earlier in this lecture): 16 identities $\times$ 4 crops per batch, margin 0.3.
- **BNNeck (batch-norm neck).** ID loss separates identities by angle, triplet loss by Euclidean distance, and on one shared feature the two conflict. BNNeck puts the triplet loss on the pooled feature f_t , maps f_t through a batch-norm (BN) layer (covered earlier) to f_i , and feeds f_i to a bias-free classifier for the ID loss. Retrieval uses f_i with cosine distance.
- **Last stride 1.** Dropping the downsampling in ResNet-50's last stage turns a 256×128 crop into a 16×8 feature map, with no new parameters.
- **Market-1501.** Six training tricks, three of them above, lift a standard baseline from 87.7% rank-1 / 74.0% mAP to 94.5% / 85.9%; BNNeck alone adds 2.1 / 4.0 points.



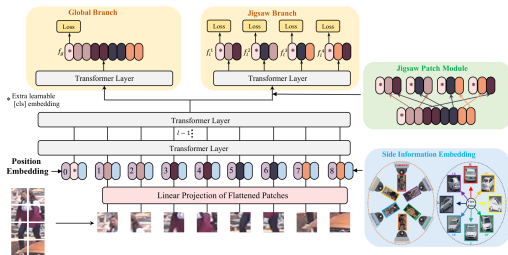
"In the inference stage, we choose f_i following the BN layer to do the retrieval." Luo et al., "Bag of Tricks and A Strong Baseline for Deep Person Re-identification," CVPRW 2019, [arXiv:1903.07071v3](https://arxiv.org/abs/1903.07071v3), Figure 5(b).

- **Backbone.** TransReID (He et al., ICCV 2021) swaps the strong baseline's ResNet-50 for ViT-B/16 (covered earlier), with ID and triplet losses on the output class token.
- **Side information embedding (SIE).** A learned vector for the image's camera (and viewpoint, when labelled) is added to every input token:

$$Z'_0 = Z_0 + \lambda S_C[r]$$

Z_0 : patch plus position embeddings; $S_C[r]$: embedding of camera r , shared by all patches of the image; λ : its weight. The aim is to reduce camera-specific bias.

- **Jigsaw patch module (JPM).** The last transformer layer is split into two independent copies. One outputs the global feature f_g ; the other shifts and shuffles the patch tokens into $k = 4$ groups, and each group plus the class token yields a local feature f_l^j drawn from the whole body. The test feature concatenates f_g and all f_l^j .
- **MSMT17 mAP** (Table 5): ViT baseline 61.0, +JPM 63.6, +SIE 62.4, both 64.9. With overlapping patches (sliding window): 67.4 mAP / 85.3 rank-1, and 69.4 / 86.2 at 384×128 (Table 6); CNNs in that table reach at most 60.8 mAP.



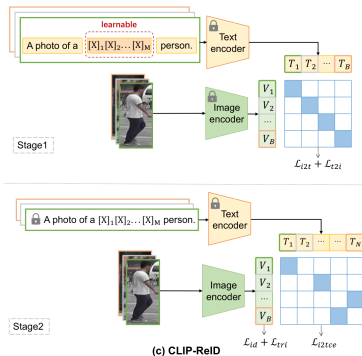
"Side Information Embedding (light blue) encodes non-visual information such as camera or viewpoint into embedding representations." He et al., "TransReID: Transformer-Based Object Re-Identification," ICCV 2021, [OpenAccess](#), Figure 4.

- **The gap.** CLIP (covered earlier) classifies by matching an image to prompts such as “a photo of a {label}”. Re-ID labels are bare indexes, so there is no text to encode.
- **Stage 1.** CLIP-ReID (Li et al., AAAI 2023) gives each training identity $M = 4$ learnable tokens in “A photo of a $[X]_1[X]_2 \dots [X]_M$ person,” like CoOp’s context vectors (earlier in this lecture) but one set per identity. Both CLIP encoders are frozen; only the tokens are trained, with image-to-text and text-to-image contrastive losses in the batch. Each identity y ends with a fixed text feature T_y .
- **Stage 2.** With the text side frozen, the image encoder is fine-tuned with ID loss, triplet loss and an image-to-text cross-entropy (i2tce) over all N identities:

$$\mathcal{L}_{i2tce}(i) = - \sum_{k=1}^N q_k \log \frac{\exp(s(V_i, T_{y_k}))}{\sum_{a=1}^N \exp(s(V_i, T_{y_a}))}$$

V_i : feature of image i ; s : dot product of projected features; q_k : label-smoothed target. The T_y act as fixed class prototypes.

- **MSMT17, ViT-B/16** (Tables 2 and 5): CLIP’s image encoder with the usual re-ID losses, 66.1 mAP; tokens and encoder trained together in one stage, 68.9; two stages, 73.4 mAP / 88.7 rank-1, or 75.8 / 89.7 with SIE and overlapping patches (TransReID with both: 67.4 / 85.3). Market-1501: 89.6 / 95.5.



Locks mark frozen parts. Li et al., “CLIP-ReID: Exploiting Vision-Language Model for Image Re-Identification without Concrete Text Labels,” AAAI 2023, [arXiv:2211.13977v4](https://arxiv.org/abs/2211.13977v4), Figure 2(c).

- **LUPerson** (Fu et al., CVPR 2021): 4,180,243 unlabelled person crops detected in YouTube street-view videos, from an estimated 219,848 people. Contrastive pretraining on it with MoCo v2 (MoCo covered earlier) shows colour jitter to be harmful, since colour is a main cue for telling people apart.
- **SOLIDER** (semantic controllable self-supervised learning; Chen et al., CVPR 2023) starts from DINO (covered earlier), whose features group a person's parts by appearance. It clusters token features with k-means (covered earlier) into background and three parts ordered top to bottom (upper body, lower body, shoes) and adds a token-level classification loss on these pseudo-labels.
- **Semantic controller.** A scalar $\lambda \in [0, 1]$ becomes a scale and a shift on the features after every Swin block (covered earlier) and weights the part loss:

$$\mathcal{L} = \alpha \mathcal{L}_{\text{dino}}(F(\lambda)) + \lambda(1 - \alpha) \mathcal{L}_{\text{sm}}(F(\lambda))$$

$F(\lambda)$: features under λ ; $\mathcal{L}_{\text{dino}}$: DINO loss; $\mathcal{L}_{\text{sm}}$: part loss; $\alpha = 0.5$. Re-ID fine-tunes with $\lambda = 0.0$ – 0.2 (appearance), parsing and detection with 0.8 – 1.0 .

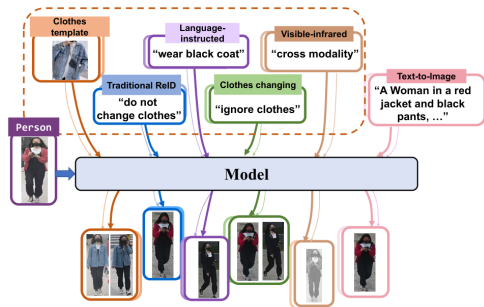
Fine-tuned with TransReID on Swin-T (tiny), mAP / rank-1:

Pretraining	MSMT17	Market-1501
ImageNet, supervised	49.7 / 73.6	78.1 / 90.2
LUP1M, DINO	63.3 / 83.2	89.6 / 95.9
+ part labels	61.6 / 82.2	89.5 / 95.5
+ controller	63.9 / 83.8	89.9 / 96.1

Last row: SOLIDER. LUP1M: a random 1M-image subset of LUPerson, close to ImageNet in size. Chen et al., "Beyond Appearance: a Semantic Controllable Self-Supervised Learning Framework for Human-Centric Visual Tasks," CVPR 2023, [arXiv:2303.17602v1](https://arxiv.org/abs/2303.17602v1), Table 1.

- Self-supervised pretraining on person crops instead of supervised ImageNet adds 13.6 mAP on MSMT17; the authors credit LUPerson's street scenes, which resemble re-ID data. Part labels alone cost 1.7 mAP; with the controller, SOLIDER ends 0.6 above plain DINO.
- SOLIDER on Swin-B (base): 77.1 mAP / 90.7 rank-1 on MSMT17, 93.9 / 96.9 on Market-1501 (Table 2).

- **Clothes-changing re-ID.** Most re-ID work assumes people keep their clothes, and models overfit to clothing. The clothes-based adversarial loss (CAL; Gu et al., CVPR 2022) alternates, on colour (RGB) images only: fit a clothes classifier on the backbone features, then train the backbone so that this classifier cannot tell one identity's outfits apart. On the PRCC benchmark CAL reaches 100% rank-1 with the same clothes and 55.2% with changed clothes (Table 2).
- **Animal re-ID.** Ecologists identify individual animals from markings. MegaDescriptor (Cermak et al., WACV 2024) is a Swin-L (large) trained with ArcFace (earlier in this lecture) on 29 public animal datasets; a query takes the identity of its nearest reference image by cosine similarity, and every query animal is in the reference set. On the HumpbackWhaleID dataset, whose reference images were part of its training data, it is right for 77.81% of queries, off-the-shelf DINOv2 (ViT-L) features for 6.44% (Table 4).
- **Instruction-driven re-ID.** Instruct-ReID (He et al., CVPR 2024) handles six re-ID variants as one task by attaching an instruction to the query: a phrase such as "Ignore clothes", a clothing image, or sentences describing the target. With sentences (COCAS+ Real2 test set), mAP goes from 14.9 for a ViT-B that ignores the text to 39.8 for their multi-task model (Table 4).



(a) One model for diverse ReID

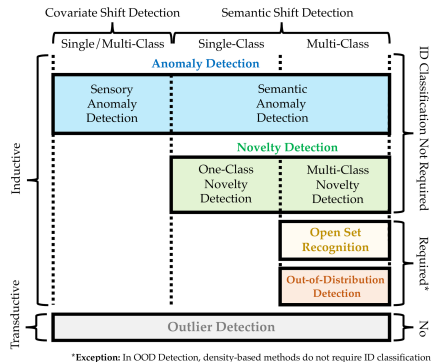
Six re-ID tasks expressed as instructions to one model. He et al., "Instruct-ReID: A Multi-purpose Person Re-identification Task with Instructions," CVPR 2024, [OpenAccess](#), Figure 1(a).

- ▶ **A surveillance technology.** Re-ID “aims to search the target person from surveillance videos across different locations and times” (Gu et al., CVPR 2022). Matches across cameras and hours add up to a record of where a person went, and errors are routine: SOLIDER’s 90.7% rank-1 on MSMT17 still ranks a different person first for 9.3% of queries.
- ▶ **Data with a history.** DukeMTMC, the multi-camera tracking dataset from which the DukeMTMC-reID benchmark was cut, was retracted by its creators in April 2019 after an investigation raised ethical concerns about how the data was collected and used. Copies kept circulating through derived datasets: a 20% sample of citing papers contains 73 uses in 2020. LUPerson was cut from crawled YouTube videos, and its paper describes no consent or face-blurring step.

Practice	What it means for a re-ID system
Decide whether to build it	State who is searched, by whom, and what follows a match; stop if the answer is open-ended tracking of people.
Fixed scope	Fix the task, the cameras and the time window; reject queries outside them.
Access control	Restrict who can query; log every query with its reason, and review the log.
Deletion schedule	Delete crops and embeddings after a fixed period; keep only needed aggregates.
Opt-in where possible	Prefer settings where people agree to take part (a team studying its own players) and wildlife.
Human review	No automatic action on a match; a person checks each result first.
Dataset stewardship	Terms of use that cover trained models and derivatives; retraction notices on all copies.

Retraction facts and the stewardship row: Peng et al., “Mitigating Dataset Harms Requires Stewardship: Lessons from 1000 Papers,” NeurIPS 2021 Datasets and Benchmarks Track, [arXiv:2108.02922v2](https://arxiv.org/abs/2108.02922v2), Figure 1, Table 2 and Section 8.1. The other rows are course guidance.

- ▶ **Closed world:** a classifier trained on K known classes assumes every test image is **in-distribution (ID)**; shown an unseen class, it still picks one of its K labels.
- ▶ **Open-set recognition (OSR)** (Scheirer et al., TPAMI 2013): classify the known classes and reject the unknown ones; “the classifier must also support rejection.”
- ▶ **Out-of-distribution (OOD) detection:** flag inputs whose class is outside the training label set (**semantic shift**). Yang et al. (IJCV 2024) treat it and OSR as one problem.
- ▶ Related tasks: anomaly, novelty and outlier detection. Autoencoder reconstruction error (covered earlier) is an anomaly score.
- ▶ **Unlike the distribution shift earlier in this lecture:** ImageNet-R and ImageNet-Sketch are covariate shift (covered earlier: same classes, new style), scored by accuracy; under semantic shift the right answer is *unknown*.



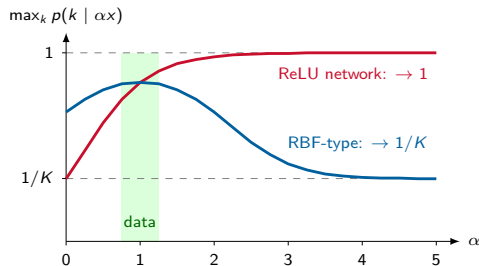
Tasks sorted by shift type, number of ID classes, and whether ID classification is required. Yang et al., “Generalized Out-of-Distribution Detection: A Survey,” IJCV 2024, [arXiv:2110.11334v3](https://arxiv.org/abs/2110.11334v3), Figure 1.

- ▶ **Maximum softmax probability (MSP)** baseline (Hendrycks and Gimpel, ICLR 2017): score $s(x) = \max_k p(k | x)$, with $p(k | x)$ the softmax output for class k . Correctly classified images tend to score higher than misclassified and OOD images.
- ▶ The value alone misleads: their MNIST classifier gives Gaussian noise a mean MSP of 91%.
- ▶ **Cause** (Hein et al., CVPR 2019): a ReLU network is piecewise affine, and its outer linear regions extend to infinity. For almost every input x ,

$$\lim_{\alpha \rightarrow \infty} \max_k \frac{e^{z_k(\alpha x)}}{\sum_{j=1}^K e^{z_j(\alpha x)}} = 1,$$

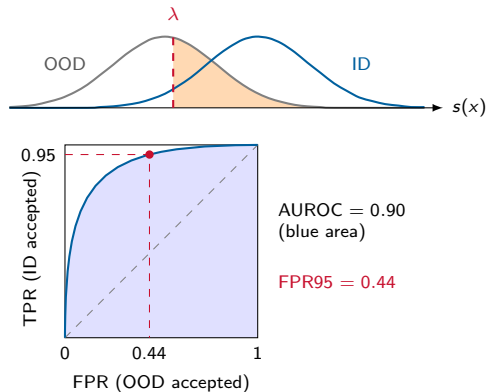
where z_k is the logit of class k and $\alpha > 0$ scales the input away from the data.

- ▶ Softmax temperature (covered earlier) cannot undo this. Uniform noise scaled by a median $\alpha = 8.1$ already gets 99.9% confidence from their CIFAR-10 model.



Toy model with $K = 3$ along the ray αx : affine logits $\alpha(3, 1, 0)$, as inside one linear region of a ReLU network (red), and radial-basis-function (RBF) logits $e^{-(\alpha-1)^2/2}(3, 1, 0)$ that decay away from the data (blue). Both give 0.84 at the training point $\alpha = 1$; far away the ReLU confidence goes to 1, while the RBF-type confidence returns to $1/K$, as in Hein et al.'s Theorem 3.2.

- ▶ **Detector:** score $s(x)$ (higher for ID), threshold λ : predict a class if $s(x) \geq \lambda$, else *unknown*; λ is set on ID data, e.g. to accept 95% of ID images.
- ▶ **Convention: ID is positive.** TPR is the share of ID images accepted, FPR of OOD images accepted (TPR and FPR covered earlier); the ROC curve plots TPR against FPR as λ sweeps.
- ▶ **AUROC:** area under the ROC curve: the probability that a random ID image outscores a random OOD image (50% is chance).
- ▶ **FPR95:** the FPR at the λ that accepts 95% of ID images; lower is better.
- ▶ **AUPR-In/Out:** area under the precision-recall curve, ID or OOD positive; depends on the ID:OOD ratio.
- ▶ **OSCR curve** (open-set classification rate; Dhamija et al., NeurIPS 2018): known-class images accepted and correctly classified, against FPR.

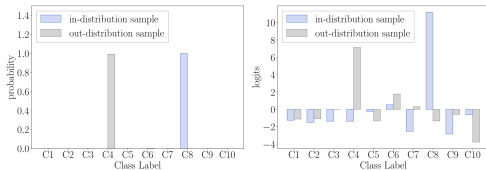


- ▶ Notation: penultimate features $h(x) \in \mathbb{R}^D$, logits $z(x) = Wh(x) + b \in \mathbb{R}^K$; no retraining.
- ▶ **MaxLogit** (maximum logit score, MLS; Hendrycks et al., ICML 2022): $s(x) = \max_k z_k(x)$. With many similar classes the softmax spreads its mass and MSP rejects hard ID images; ResNet-50 on ImageNet: FPR95 44.2 (MSP) to 35.8.
- ▶ **Energy** (Liu et al., NeurIPS 2020), with temperature T (they use $T = 1$):

$$E(x) = -T \log \sum_{k=1}^K e^{z_k(x)/T}, \quad s(x) = -E(x).$$

At $T = 1$, $\max_k z_k \leq -E \leq \max_k z_k + \log K$: a smooth MaxLogit.

- ▶ **Why MSP is biased** ($T = 1$):
 $\log \max_k p(k | x) = \max_k z_k(x) + E(x)$; ID inputs raise the first term and lower the second, so they partly cancel. CIFAR-10 WideResNet (a wider, shallower ResNet), six OOD sets: FPR95 51.04 (MSP) to 33.01 (energy).



(a) softmax scores 1.0 vs. 0.99

(b) negative energy scores: 11.19 vs. 7.11

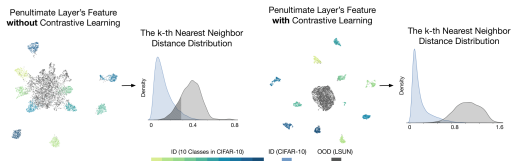
One CIFAR-10 image (ID) and one SVHN street-number image (OOD) on a CIFAR-10 WideResNet: MSP 1.0 vs 0.99, negative energy 11.19 vs 7.11. Liu et al., “Energy-based Out-of-distribution Detection,” NeurIPS 2020, [arXiv:2010.03759v4](https://arxiv.org/abs/2010.03759v4), Figure 2.

- **Mahalanobis** (Lee et al., NeurIPS 2018): class means μ_c and one shared covariance Σ of the training features $h(x)$, then

$$s(x) = \max_c -(h(x) - \mu_c)^\top \Sigma^{-1} (h(x) - \mu_c).$$

It assumes Gaussian classes: weak on ResNet-50 (table), AUROC 96.02 on ViT-B/16 (other OOD sets; Wang et al., CVPR 2022).

- **Deep k-NN** (Sun et al., ICML 2022): $s(x) = -\|u - u_{(k)}\|_2$, with $u = h(x)/\|h(x)\|_2$ and $u_{(k)}$ its k -th nearest normalised training feature. No distribution assumption; the threshold uses ID data only.
- Approximate nearest-neighbour search (earlier in this lecture) keeps k -NN fast. On SupCon features (earlier in this lecture), which form compact classes, its FPR95 drops from 54.68 to 38.47.



CIFAR-10 (colours) vs LSUN scene images (grey), without and with contrastive training. Sun et al., “Out-of-Distribution Detection with Deep Nearest Neighbors,” ICML 2022, [arXiv:2204.06507v3](https://arxiv.org/abs/2204.06507v3), Figure 1.

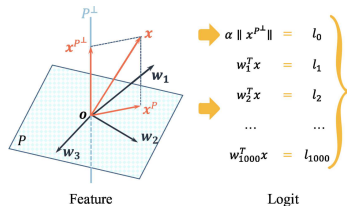
Score	Training	FPR95 ↓	AUROC ↑
MSP	CE	66.95	81.99
Mahalanobis	CE	87.43	55.47
k -NN	CE	54.68	84.37
k -NN	SupCon	38.47	90.91

ResNet-50 on ImageNet-1K, cross-entropy (CE) or SupCon training, $k = 1000$; mean over iNaturalist, SUN, Places, Textures. Sun et al., Table 4.

- Feature scores ignore class identity; logit scores miss feature directions outside the span of the class weights. **Virtual-logit Matching (ViM)** (Wang et al., CVPR 2022) uses both.
- Figure notation: x is the penultimate feature, shifted so the logits $l_i = w_i^\top x$ need no bias; C classes.
- P : span of the top- D eigenvectors of $X^\top X$ for training features X (PCA-style, covered earlier); the residual $x^{P^\perp}$ is the part of x outside P .
- Virtual logit $l_0 = \alpha \|x^{P^\perp}\|$, with α set so its training mean equals the mean maximum logit:

$$\text{ViM}(x) = \frac{e^{l_0}}{e^{l_0} + \sum_{i=1}^C e^{l_i}} \quad (\text{OOD if large}).$$

Ranking by ViM equals ranking by $l_0 + E(x)$: the energy plus the scaled residual norm.



Feature and logit stages. Wang et al., “ViM: Out-Of-Distribution with Virtual-logit Matching,” CVPR 2022, [arXiv:2203.10807v1](https://arxiv.org/abs/2203.10807v1), Figure 3 (centre).

Score	BiT	ViT-B/16
MSP	77.25 / 77.83	89.40 / 41.65
Energy	78.48 / 79.68	94.90 / 22.43
Mahalanobis	86.62 / 53.34	96.02 / 19.45
ViM	90.91 / 41.46	96.23 / 18.07

AUROC / FPR95 on ImageNet-1K, averaged over OpenImage-O, Textures, iNaturalist and ImageNet-O. BiT (Big Transfer): a ResNet-v2 trained on ImageNet-1K; the ViT is pre-trained on ImageNet-21K. Wang et al., Table 2.

- ▶ All three edit the penultimate activations $h(x) \in \mathbb{R}^D$ (non-negative after the ReLU) before the last layer, then score with the energy $-E(x)$; no retraining.
- ▶ **ReAct** (rectified activations; Sun et al., NeurIPS 2021): $\bar{h} = \min(h(x), c)$ per unit, c the 90th percentile of ID activations. OOD inputs push a few units to outsized values; clipping removes them. ID top-1: 75.08% to 73.75%.
- ▶ **ASH-S** (activation shaping; Djuricic et al., ICLR 2023): per image, zero activations below the image's 90th percentile, multiply the rest by e^{s_1/s_2} (s_1, s_2 : sums before and after pruning). ID top-1: 76.13% to 74.98%.
- ▶ **SCALE** (Xu et al., ICLR 2024): keep every activation and multiply all by e^r , $r = Q(h)/Q_p(h)$ (sum of all activations over the sum above the p -th percentile). The arg max, hence ID accuracy, is unchanged; they find ASH's gains come from scaling, while pruning hurts.

Score	FPR95 ↓	AUROC ↑
MSP	66.95	81.99
Energy	58.41	86.17
ReAct	31.43	92.95
ASH-S	22.80	95.12
SCALE	20.05	95.71

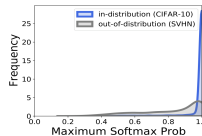
ResNet-50 on ImageNet-1K, averaged over iNaturalist, SUN, Places365 and Textures. Xu et al., "Scaling for Training Time and Post-hoc Out-of-distribution Detection Enhancement," ICLR 2024, proceedings.iclr.cc, Table 7.

On harder OOD classes that sit close to ImageNet's (covered later in this lecture), a ResNet-50 with SCALE still has FPR95 59.76 (their Table 2).

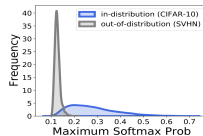
- **Outlier Exposure (OE)** (Hendrycks et al., ICLR 2019): train on auxiliary outlier images x' , disjoint from the test OOD sets, with an extra loss weighted 0.5: the cross-entropy from $p(\cdot | x')$ to the uniform distribution. OSR protocols discourage such extra data by design (Yang et al.).
- **LogitNorm** (logit normalisation; Wei et al., ICML 2022): cross-entropy keeps inflating the logit norm $\|z\|$ on images already classified correctly, so confidence grows for ID and OOD alike. Train on normalised logits:

$$\mathcal{L} = -\log \frac{e^{z_y / (\tau \|z\|)}}{\sum_{k=1}^K e^{z_k / (\tau \|z\|)}},$$

with label y and temperature τ (0.04 on CIFAR-10). CIFAR-10, MSP score, six OOD sets: FPR95 49.52 to 15.65 (WRN-40-2, a WideResNet); test accuracy 95.14% vs 95.01% (ResNet-34).



(a) Cross-entropy Loss



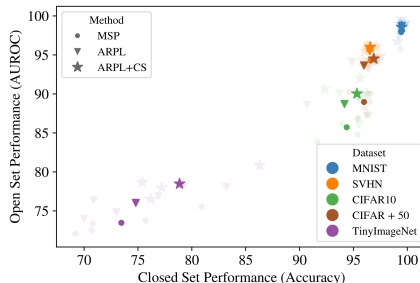
(b) LogitNorm Loss (Ours)

MSP histograms, CIFAR-10 (ID) vs SVHN (OOD), after cross-entropy (a) and LogitNorm (b) training. Wei et al., “Mitigating Neural Network Overconfidence with Logit Normalization,” ICML 2022, [arXiv:2205.09310v2](#), Figure 4.

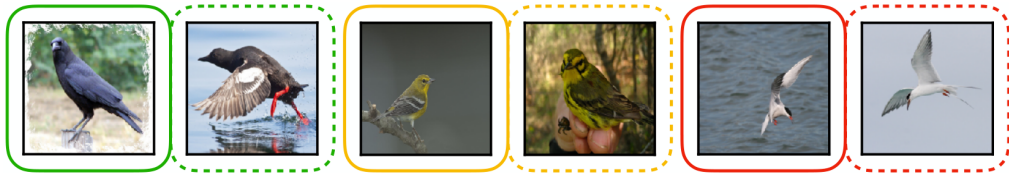
ID data	MSP	MSP after OE
CIFAR-10	34.9	9.5
CIFAR-100	62.7	38.5
Places365	63.5	28.2

FPR95 before and after OE fine-tuning. Hendrycks et al., “Deep Anomaly Detection with Outlier Exposure,” ICLR 2019, [arXiv:1812.04606v3](#), Table 1.

- ▶ **Finding** (Vaze et al., ICLR 2022): on standard open-set benchmarks, AUROC grows almost linearly with closed-set accuracy, Pearson $\rho = 0.95$ across MSP, ARPL (Adversarial Reciprocal Point Learning) and ARPL+CS (ARPL with generated confusing samples).
- ▶ **ImageNet scale:** across CNN, ViT and MLP-Mixer (all-MLP) classifiers, $\rho = 0.88$ on the Hard unknown split and 0.63 on the Easy one (next slide); within ResNets, 0.99 and 1.00. ViT-B/16 lies above the trend, but its ImageNet-21K pre-training contains the “unknown” classes.
- ▶ **Proposed reason:** cross-entropy is a proper scoring rule (optimal at the true probabilities), so lower test error should bring better calibration (confidence matching accuracy), and low confidence then flags a new class more reliably.
- ▶ **Recipe:** longer training, stronger augmentation, label smoothing, then MLS scoring (both earlier in this lecture). TinyImageNet AUROC: 57.7 for the published MSP baseline, 83.0 for MLS; ARPL+CS: 78.2 published, 82.5 retrained. Over six benchmarks MLS gains 15.6 points on average, 0.03 from the retrained ARPL+CS.



Bold markers: mean over five known/unknown class splits; faint markers: single splits. Vaze et al., “Open-Set Recognition: a Good Closed-Set Classifier is All You Need?” ICLR 2022, [arXiv:2110.06207v2](https://arxiv.org/abs/2110.06207v2), Figure 2.



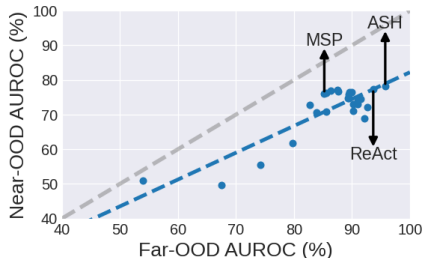
Unknown CUB class (dashed) next to its closest known class (solid); green, orange, red: Easy, Medium, Hard. Vaze et al., [arXiv:2110.06207v2](https://arxiv.org/abs/2110.06207v2), Figure 5.

- **Semantic Shift Benchmark (SSB):** unknown classes of CUB, Stanford Cars and FGVC-Aircraft (earlier in this lecture) are binned Easy, Medium or Hard by attribute similarity to the known classes (CUB: 312 attributes, e.g. `has_bill_shape::hooked`).
- **ImageNet:** the 1K classes are known; ImageNet-21K-P classes sorted by summed hierarchy distance to them form 1,000-class Easy and Hard sets.
- Old splits (6 known, 4 unknown CIFAR-10 classes) leave novelty undefined. Hard costs MLS 5.9 AUROC points on ImageNet; 10× more random unknowns cost about 0.6.

Dataset	Known classes	Closed acc.	AUROC	
			Easy	Hard
CUB	100	86.2	88.3	79.3
Stanford Cars	98	97.1	94.0	82.2
FGVC-Aircraft	50	91.7	90.7	82.3
ImageNet	1000	78.8	78.7	72.8

ResNet-50 scored with MLS; Hard includes the Medium bin where one exists. Vaze et al., [arXiv:2110.06207v2](https://arxiv.org/abs/2110.06207v2), Tables 2–3.

- ▶ **Setup** (Zhang et al., DMLR 2024): ImageNet-1K is the in-distribution (ID) set. Near-OOD: SSB-hard (980 classes from SSB's 1,000-class Hard ImageNet split) and NINCO (5,879 hand-curated images). Far-OOD: iNaturalist, Textures, OpenImage-O. Nearly 40 methods, tuned on held-out validation sets.
- ▶ **Near-OOD stays hard:** with a ResNet-50, ASH gets 95.74 AUROC on far-OOD but 78.17 on near-OOD (MSP: 85.23 and 76.02); near-OOD gains trail far-OOD gains (right).
- ▶ **No single winner:** ReAct and ASH lead on ImageNet but are less competitive on CIFAR. AugMix training (mixed augmented copies of each image) plus ASH gives the best near-OOD AUROC, 82.16.
- ▶ **Backbones:** ViT-B/16 and Swin-T detect OOD no better than ResNet-50, despite higher top-1 accuracy (81.1% and 81.6% vs 76.2%). A DINOv2 linear probe beats a supervised ViT-B on every OOD split; zero-shot CLIP helps mainly on far-OOD.
- ▶ **Full-spectrum test:** adding covariate-shifted ID images (ImageNet-C corruptions, -R, -V2) costs most methods over 10 near-OOD AUROC points: detectors also reject shifted images of known classes.



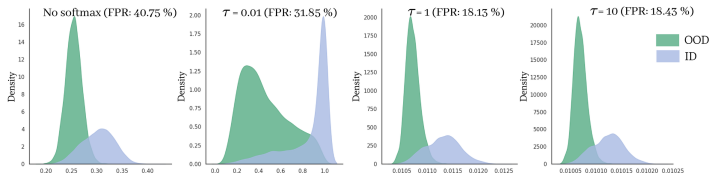
One dot per method on ImageNet-1K with a ResNet-50; grey dashes: equal near- and far-OOD AUROC; blue dashes: fitted trend. Zhang et al., "OpenOOD v1.5: Enhanced Benchmark for Out-of-Distribution Detection," DMLR 2024, [arXiv:2306.09301v5](https://arxiv.org/abs/2306.09301v5), Figure 2.

- **Zero-shot OOD detection:** only the K ID class names are given, no ID images. MCM (Maximum Concept Matching; Ming et al., NeurIPS 2022) scores an image x with CLIP (covered earlier):

$$S_{\text{MCM}}(x) = \max_i \frac{e^{s_i(x)/\tau}}{\sum_{j=1}^K e^{s_j(x)/\tau}}$$

$s_i(x)$: cosine between the image embedding of x and the text embedding of “this is a photo of a $\langle y_i \rangle$ ”; τ : temperature (1 in the paper). Reject x if $S_{\text{MCM}}(x) < \lambda$, with λ passing 95% of ID images.

- **Role of the softmax:** OOD images have near-equal cosines to all ID labels, and with many classes an ID image’s top cosine is only slightly higher; a softmax with moderate τ widens the gap (provably, above a threshold τ , under this near-uniformity).
- **Temperature (below):** FPR95 is 40.75% without softmax, 31.85% at $\tau = 0.01$ (CLIP’s smallest training temperature), 18.13% at $\tau = 1$; the predicted class is unchanged. ImageNet-1K, ViT-B/16: AUROC 90.77, FPR95 42.74 (raw maximum cosine: 83.59, 69.08).



ImageNet-100 as ID, iNaturalist as OOD. Ming et al., “Delving into Out-of-Distribution Detection with Vision-Language Representations,” NeurIPS 2022, [arXiv:2211.13445v1](https://arxiv.org/abs/2211.13445), Figure 4.

- **Idea** (Jiang et al., ICLR 2024): MCM compares an image only with the K ID labels. NegLabel adds $M = 10,000$ negative labels, nouns and adjectives from the WordNet lexical database whose text embeddings are far from every ID label; OOD images match some of them more closely than ID images do.
- **NegMining**: rank candidate words by the 5th percentile of their negative cosine similarities to the ID labels (a minimum that tolerates outlier labels) and keep the top M .

- **Score**:

$$S(x) = \frac{\sum_{i=1}^K e^{c_i/\tau}}{\sum_{i=1}^K e^{c_i/\tau} + \sum_{j=1}^M e^{\tilde{c}_j/\tau}}$$

c_i , $\tilde{c}_j$: cosine similarity of the image embedding of x with ID label i and negative label j ; $\tau = 0.01$. The sum over ID labels tolerates confusions between ID classes (husky vs malamute); averaging over 100 groups of negatives limits chance matches.

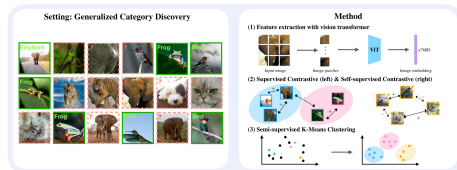
- **Cost**: about 1 ms per image at $M = 10,000$; ID predictions are unchanged.

Method	Training	AUROC $\uparrow$	FPR95 $\downarrow$
MSP	fine-tuned	81.63	69.61
Energy	fine-tuned	91.54	39.89
k -NN	fine-tuned	90.97	42.19
MCM (re-run)	none	90.82	43.93
NegLabel	none	94.21	25.40

ImageNet-1K as ID, CLIP ViT-B/16 (fine-tuned on ImageNet-1K where marked), mean of four OOD sets; MCM as re-run by Jiang et al. Jiang et al., "Negative Label Guided OOD Detection with Pretrained Vision-Language Models," ICLR 2024, [arXiv:2403.20078v1](https://arxiv.org/abs/2403.20078v1), Table 1.

- **LoCoOp** (Miyai et al., NeurIPS 2023) trains CoOp prompts (earlier in this lecture) on a few labelled images and pushes patch features whose top-200 classes miss the true label, mostly background, toward uniform ID predictions. 16-shot AUROC with MCM: 92.69 (CoOp: 91.93).

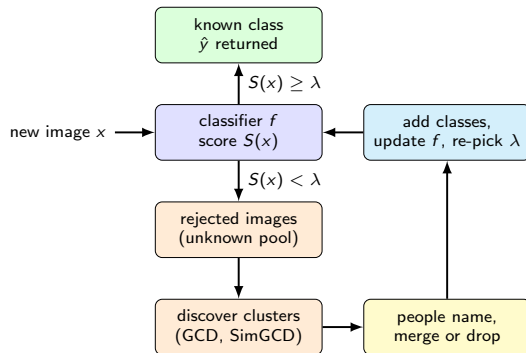
- **Setting** (GCD: Generalized Category Discovery; Vaze et al., CVPR 2022): labelled images $\mathcal{D}_L$ of classes $\mathcal{Y}_L$, unlabelled images $\mathcal{D}_U$ of classes $\mathcal{Y}_U \supset \mathcal{Y}_L$; every image in $\mathcal{D}_U$ gets an old class or a new cluster.
- **GCD**: fine-tune a DINO ViT-B/16 (covered earlier) with SupCon (earlier in this lecture) on $\mathcal{D}_L$ plus self-supervised contrastive learning on all images, then run k -means with each labelled image fixed to its class's cluster.
- **Metric**: clustering accuracy (ACC), the fraction of $\mathcal{D}_U$ labelled correctly under the best one-to-one cluster-to-class map, found once by the Hungarian algorithm (an exact solver for one-to-one assignment); reported on All, Old ($\mathcal{Y}_L$) and New classes.
- **SimGCD** (Wen et al., ICCV 2023): cosine prototypes on backbone features, trained toward the sharpened prediction of another view (self-distillation); $-\varepsilon H(\bar{\mathbf{p}})$, the weighted entropy of the batch-mean prediction $\bar{\mathbf{p}}$, counters the bias toward Old classes. CUB inference: 18s vs 9 min for k -means.



Vaze et al., "Generalized Category Discovery," CVPR 2022, [arXiv:2201.02609v2](https://arxiv.org/abs/2201.02609v2), Figure 1.

	GCD		SimGCD	
ACC (%)	All	New	All	New
CUB	51.3	48.7	60.3	57.7
Stanford Cars	39.0	29.9	53.8	45.0
FGVC-Aircraft	45.0	46.9	54.2	51.8

SSB splits; k -means on raw DINO features: 34.3 All on CUB. Wen et al., "Parametric Classification for Generalized Category Discovery: A Baseline Study," ICCV 2023, [arXiv:2211.11727v4](https://arxiv.org/abs/2211.11727v4), Tables 2 and 5.



Adapted from Bendale and Boulton, "Towards Open World Recognition," CVPR 2015, [arXiv:1412.5687v1](https://arxiv.org/abs/1412.5687v1), Figure 1, with a discovery step added before labelling.

- **Open-world recognition** (Bendale and Boulton, CVPR 2015): detect unknowns, collect them, have people label them, then learn the new classes incrementally. Category discovery reduces the labelling to naming clusters.
- **Threshold:** pick λ on held-out ID images for a target TPR (e.g. 95%), using validation data only, and re-pick it after every update of f , since the score distribution moves with the model.
- **Monitor the rejection rate:** at 95% TPR, about 5% of ID images are rejected by design. A sustained rise signals new classes or covariate shift; inspect samples first, because detectors also reject shifted images of known classes.
- **Before adding classes:** wait for stable clusters, estimate how many there are (GCD's estimate for CUB: 231 against 200 true classes), and re-test old classes on a fixed audit set after each update.

- ▶ **Benchmarks leak and carry label noise.** GLDV2-clean contains the landmarks of 38 of the 70 ROxford queries, and removing them costs published methods 10.6 to 29.7 mAP on ROxford Hard. DINOv2's training data includes images retrieved around Cars, Aircraft, CUB, ROxford and RParis. Experts found about 4% wrong labels across CUB-200-2011, test images included, and Van Horn et al. (CVPR 2015) conclude that fine-grained systems need an expert-curated test set.
- ▶ **Reported gains depend on the protocol.** Under one backbone, embedding size and tuning budget, metric-learning losses from 2006 to 2019 land within a few MAP@R points; new backbones and pretraining data moved Recall@1 far more.
- ▶ **Adaptation still trades something.** End-to-end fine-tuning lowers CLIP's accuracy under distribution shift (73.4 to 68.6 on average); weight-space ensembling recovers it, but learned prompts still lose accuracy on new classes (CoOp 63.22 against 74.22 for zero-shot CLIP).
- ▶ **Unknowns close to the known classes stay hard.** Near-OOD AUROC trails far-OOD AUROC (ASH: 78.17 against 95.74), the Hard semantic-shift split costs 5.9 AUROC points on ImageNet, and mixing in covariate-shifted images of known classes costs most detectors over 10 points.
- ▶ **Zero-shot rejection is only as good as the label text.** MCM and NegLabel compare each image only with text labels; with fine-grained labels that CLIP separates poorly (ViT-L/14@336px: 37.2% zero-shot on FGVC-Aircraft), the text side is a weak basis for rejecting look-alike unknowns.
- ▶ **Thresholds and identities need maintenance.** A threshold chosen at 95% TPR must be re-picked after every model update and watched under shift; re-identification tracks people, and retracted datasets keep circulating through derived copies.

Numbers from the earlier slides of this lecture, where each is credited; full entries in the references.

- ▶ **Adapt a pretrained classifier and keep its accuracy under shift.** Fine-tuning CLIP ViT-L end-to-end lifts ImageNet top-1 from 76.6 to 86.2 but drops mean shift accuracy from 73.4 to 68.6; WiSE-FT at $\alpha = 0.5$ reaches 86.8 and 76.9. LP-FT and greedy soups are other cheap fixes; CoOp, CoCoOp and Tip-Adapter adapt from 16 images per class.
- ▶ **Fine-grained recognition is limited by supervision as much as by architecture.** Look-alike classes need expert labels (CUB: about 4% label errors). TransFG reaches 91.7% on CUB and a frozen DINOv2 g/14 probe 91.6%; BioCLIP (TreeOfLife-10M, taxonomic names) lifts mean zero-shot accuracy on 10 biology tasks from 21.9 (CLIP) to 39.4; vision-language assistants trail CLIP until trained on classification data.
- ▶ **Metric learning trains a distance and is judged by ranking.** Pair and triplet losses need mining; SupCon, proxies and ArcFace avoid it. Under one fair protocol (MAP@R), losses from 2006 to 2019 differ by a few points, while stronger pretrained ViTs lifted CUB Recall@1 from 68.4 (Proxy Anchor) to 88.8 (Unicom).
- ▶ **Retrieval is a two-stage system.** A global descriptor (GeM, or frozen DINOv2 at 54.0 mAP on ROxford Hard) ranks the gallery through an IVFADC or HNSW index; query expansion or SuperGlobal re-ranks the shortlist. Out of domain (ILIAS), SigLIP with one trained linear layer reaches 28.9 mAP@1k; DreamSim matches human judgments on 96.2% of triplets.
- ▶ **Re-identification is retrieval of unseen identities across cameras,** scored by CMC rank- k and mAP: the six-trick strong baseline with BNNeck reaches 94.5% rank-1 and 85.9% mAP on Market-1501, and MSMT17 mAP rose from 67.4 (TransReID) to 73.4 (CLIP-ReID) and 77.1 (SOLIDER). Deploy it only with a fixed scope, access control, a deletion schedule and human review.
- ▶ **Rejecting unknowns takes a score, a threshold and the right metric.** Softmax confidence tends to 1 far from the data; logit, feature and activation-shaping scores cut ResNet-50's ImageNet FPR95 from 66.95 (MSP) to 20.05 (SCALE). Report AUROC and FPR95 with ID as the positive class.
- ▶ **Foundation models change how unknowns are handled.** Closed-set accuracy predicts open-set AUROC (Pearson 0.95); near-OOD stays hard (ASH: 78.17 against 95.74 AUROC on far-OOD); CLIP rejects unknowns from text labels alone (NegLabel: 94.21 AUROC); SimGCD turns rejected images into new classes (60.3% clustering accuracy on CUB).

1. Wortsman et al., "Robust fine-tuning of zero-shot models," CVPR 2022. [arXiv:2109.01903v3](#) (source of Figures 1 and 2)
2. Kumar et al., "Fine-Tuning can Distort Pretrained Features and Underperform Out-of-Distribution," ICLR 2022. [arXiv:2202.10054v1](#) (source of Figure 1)
3. Wortsman et al., "Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time," ICML 2022. [arXiv:2203.05482v3](#)
4. Zhou et al., "Conditional Prompt Learning for Vision-Language Models," CVPR 2022. [arXiv:2203.05557v2](#) (source of Figure 1(b))
5. Zhou et al., "Learning to Prompt for Vision-Language Models," IJCV 2022. [arXiv:2109.01134v6](#)
6. Zhang et al., "Tip-Adapter: Training-free Adaption of CLIP for Few-shot Classification," ECCV 2022. [arXiv:2207.09519v1](#) (source of Figure 1)
7. Menon and Vondrick, "Visual Classification via Description from Large Language Models," ICLR 2023. [OpenReview](#) (source of Figure 1)
8. Pratt et al., "What does a platypus look like? Generating customized prompts for zero-shot image classification," ICCV 2023. [arXiv:2209.03320v3](#)
9. Wei et al., "Fine-Grained Image Analysis with Deep Learning: A Survey," TPAMI 2022. [arXiv:2111.06119v2](#) (source of Figures 4 and 8)
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